

CS-AML

Data Model Specification

Version 0.1.4

Approved Internal Specification Baseline

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Civil Society Anti-Money Laundering & Financial Intelligence Framework

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Civil Society Anti-Money Laundering & Financial Intelligence Framework

Version 0.1.4 | Data Specification (Approved Internal Specification Baseline)**Document status — v0.1.4**

Version: 0.1.4 — Approved Internal Specification Baseline (2026-10-10, tag v0.1.4-spec). Supersedes v0.1.3 (2026-10-09, tag v0.1.3-spec). *[v0.1.4]*

Supersedes: CS-AML Data Model Specification v0.1. The DOCX/PDF files in this repository are the unchanged v0.1 baseline (legacy); this Markdown file is the canonical source.

Validation: approved by the product owner as the internal specification baseline on 2026-10-08 (v0.1.1) and 2026-10-09 (v0.1.2 and v0.1.3) and 2026-10-10 (v0.1.4; decision register and release gates in CHANGELOG.md). The v0.1.2, v0.1.3 and v0.1.4 changes come from change requests raised while implementing increments I1-I4, I5-I7 and the post-MVP follow-ups; this is not an independent audit. Acceptance criteria in this document are targets, not evidence that tests have passed.

CS-AML is not an external standard or certification. References to FATF, Wolfsberg, PPATK, UNODC or other bodies do not imply their endorsement.

Changes in 0.1.1: see CHANGELOG.md at the repository root (audit findings A01-A16).

Changes in 0.1.2: change requests CR-I1-01...CR-I4-14 approved by the product owner on 2026-10-09 (CHANGELOG.md, section v0.1.2). Each change is tagged **[v0.1.2 · CR-xx-yy]**.

Changes in 0.1.3: change requests CR-I5-01...CR-I7-08 approved by the product owner on 2026-10-09 (CHANGELOG.md, section v0.1.3). Each change is tagged **[v0.1.3 · CR-xx-yy]**.

Changes in 0.1.4: change requests CR-N-01...CR-N-14 approved by the product owner on 2026-10-10 (CHANGELOG.md, section v0.1.4). Each change is tagged **[v0.1.4 · CR-N-xx]**.

Status

This document defines the canonical information model for CS-AML v0.1.1. It is intended to be implementation-neutral and SHALL be used as the authoritative semantic reference for database schemas, APIs, graph stores, analytical tooling, exchange formats, and audit records that claim to implement the CS-AML data model. *[v0.1.1 · A01]*

Document Control

Field	Value
Document	CS-AML Data Model Specification
Version	0.1.4 <i>[v0.1.4]</i>
Status	Approved Internal Specification Baseline (2026-10-10, tag v0.1.4-spec) <i>[v0.1.1 · A01]</i>
Applies to	CS-AML Framework v0.1.4 Expanded (CS-AML_Framework_v0.1.4_Expanded.md) and derivative implementations <i>[v0.1.1 · A01]</i>
Primary audience	Framework maintainers, investigators, data architects, software engineers, security/privacy reviewers, assurance reviewers
Normative terms	SHALL/MUST = mandatory; SHOULD = recommended; MAY = optional
Design objective	Preserve evidentiary provenance, analytical uncertainty, temporal context, and privacy constraints while enabling reusable entity-centric financial intelligence.

1. Purpose and Scope

The purpose of this specification is to define the canonical data structures, semantics, constraints, and lifecycle rules required to represent civil-society financial intelligence under CS-AML. The model is deliberately designed to separate observed information from analytical interpretation and to keep provenance, confidence, and legal/privacy handling attached to the objects they govern.

Core design axiom

Case is context. Entity and Evidence are reusable truth-bearing objects. Analytical conclusions SHALL be represented separately from source material and SHALL remain traceable to the evidence and reasoning that support them.

1.1 In scope

- Case and investigation context
- Source and evidence provenance
- Claim, fact, indicator, and analytical proposition
- Entity identity and entity resolution
- Relationships, ownership, control, and affiliation
- Assets and asset attribution
- Events and temporal state
- Direct, documented, reconstructed, and hypothetical value flows
- Typology matches and indicators
- Hypotheses, intelligence gaps, assessments, and confidence
- Intelligence products, review, dissemination, and closure
- Privacy labels, handling restrictions, audit events, and retention metadata
- Implementation mappings for relational and graph-oriented systems

1.2 Out of scope

- Bank transaction ingestion protocols
- Formal regulated-entity STR/SAR schemas
- Law-enforcement evidentiary rules for criminal prosecution
- Covert collection or unauthorized access mechanisms
- Automated determinations of criminal guilt
- Vendor-specific database design

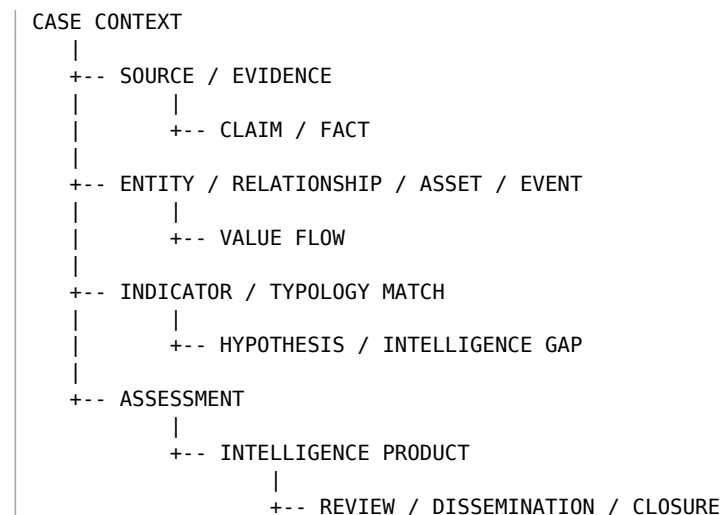
2. Normative Data Principles

ID	Principle	Normative requirement
DM-P01	Provenance by default	Every substantive evidentiary or analytical object SHALL retain machine-readable provenance to its source, creator, creation time, and applicable verification state.
DM-P02	Separation of observation and inference	Evidence, Fact, Indicator, Hypothesis, and Assessment SHALL be modeled as distinct object classes. Implementations SHALL NOT collapse them into a single notes field.
DM-P03	Temporal truth	Where a statement can change over time, the model SHALL support valid-time boundaries and SHALL avoid treating current state as historically permanent.

ID	Principle	Normative requirement
DM-P04	Entity reuse	Entities SHALL be reusable across cases. Case membership SHALL be represented as contextual association rather than by duplicating entity records per case.
DM-P05	Identity uncertainty	Entity resolution SHALL support candidate matches, confidence, merge rationale, and reversible decisions.
DM-P06	Value-flow epistemics	Direct, documented, reconstructed, and hypothetical flows SHALL be distinguishable in the data model and in any derived visualization.
DM-P07	Privacy as metadata	Sensitivity, access restrictions, retention, and dissemination limits SHALL be attached to data objects and not only to user-interface screens.
DM-P08	Immutability of originals	Original evidence SHALL be preserved. Corrections and annotations SHALL be additive or versioned; they SHALL NOT silently rewrite the historical record.
DM-P09	No guilt flag	The canonical model SHALL NOT contain a boolean field that declares a person guilty of money laundering or other crime.
DM-P10	Auditability	Material changes to identity, evidence, relationships, assessment, or dissemination state SHALL generate auditable events.

3. Canonical Information Architecture

CS-AML defines a layered information model. The layers are semantic rather than deployment-specific; one implementation MAY use PostgreSQL, a graph database, object storage, or a hybrid architecture, provided the normative semantics are preserved.



3.1 Canonical object families

Family	Canonical objects
Context	Case, CaseEntity, CaseRole, InvestigationQuestion, ScopeChange
Evidence	Source, EvidenceItem, EvidenceExtract, Claim, Fact, VerificationDecision [v0.1.1 · A10]
Knowledge graph	Entity, ResolutionDecision, PersonProfile, OrganizationProfile, AccountProfile, Asset, Relationship, Event, Location [v0.1.1 · ER]
Financial analysis	ValueFlow, ValueFlowLeg, OwnershipInterest, ControlAssertion, AssetAttribution
Analytical	Indicator, TypologyMatch, Hypothesis, HypothesisEvidenceLink, IntelligenceGap, Assessment
Product & assurance	IntelligenceProduct, Review, Dissemination, ClosureRecord
Governance	DataClassification, AccessLabel, RetentionRule, AuditEvent, VersionRecord

4. Common Object Envelope

All canonical objects SHALL expose a minimum common envelope. Implementations MAY extend this envelope but SHALL NOT redefine the meaning of the normative fields.

Field	Type	Cardinality	Meaning
id	UUID/string	Required	Globally unique immutable object identifier.
object_type	enum	Required	Canonical class name.
schema_version	string	Required	Schema version under which object was created/validated.
created_at	datetime	Required	System creation timestamp.
created_by	principal-id	Required	Human or service principal responsible for creation.
updated_at	datetime	Required	Last material update timestamp.
status	enum	Required	Lifecycle status appropriate to object class. Classes with their own lifecycle mirror it; classes without one use registry enum envelope_status (REGISTERED, RECORDED, DRAFT, FINALIZED; Annex A). [v0.1.2 · CR-12-03, CR-13-05, CR-14-09]
classification	enum	Required	Information sensitivity classification (Section 16; PUBLIC, INTERNAL, SENSITIVE, RESTRICTED, SOURCE_PROTECTED). Unknown or missing values fail closed. [v0.1.1 · A08]
access_labels	array	Conditional	Attribute-based handling labels.
case_links	array<case-id>	Conditional	Contextual case associations; not ownership. Required (1..20) when creating case-context objects (Section 7.1). [v0.1.2 · CR-12-01, CR-13-13, CR-14-12]
provenance_refs	array<id>	Conditional	Links to source/evidence/provenance objects.
confidence	object	Conditional	Confidence object where analytical uncertainty exists.
valid_from	datetime/date	Optional	Beginning of real-world validity.
valid_to	datetime/date	Optional	End of real-world validity.
record_version	integer	Required	Monotonic version number for optimistic concurrency/audit.
deleted_at	datetime	Optional	Soft-delete/tombstone marker where policy permits.

4.1 Wire enumeration convention

All controlled enumerations on the wire (database values, API payloads, exports) SHALL use UPPER_SNAKE_CASE values as listed in the field tables of this specification and in Annex A, which together form the controlled-enumeration registry. Display labels are separate, translatable presentation strings and SHALL NOT be stored or exchanged in place of wire values. Enumeration lists in other CS-AML documents derive from this registry. [v0.1.1 · A09]

5. Identifier and Naming Standard

Object identifiers SHALL be opaque and SHALL NOT encode sensitive real-world identifiers, names, or conclusions. Human-readable display codes MAY be assigned separately.

Object	Prefix example	Canonical ID	Human display code
Case	CASE	UUID	CSAML-CASE-2026-001
Entity	ENT	UUID	ENT-000145
Evidence	EVD	UUID	EVD-000381
Hypothesis	HYP	UUID	HYP-006
Assessment	ASS	UUID	ASS-003

6. Context and Case Objects

6.1 Case

Represents the bounded investigative context in which questions, scope, risks, analytical objects, and products are managed. A Case SHALL NOT own reusable entities or evidence; it SHALL link to them.

Field	Type	Req.	Cardinality	Semantics
title	string	Y	1	Neutral descriptive title.
case_code	string	Y	1	Human-readable identifier.
purpose	text	Y	1	Legitimate investigative purpose.
investigation_questions	ref[]	Y	0..n	Questions to be answered. MAY be empty while the case is DRAFT; at least one question is a precondition of case activation (gate), not of the stored record or API response. <i>[v0.1.2 · CR-I1-01]</i>
jurisdictions	code[]	N	0..n	Relevant jurisdictions.
time_scope	interval	N	0..1	Primary period under review.
lead_analyst	principal	Y	1	Accountable analyst.
risk_rating	enum	Y	1	Operational/harm risk rating; registry enum risk_rating: LOW, MEDIUM, HIGH, CRITICAL (ordered; minimal set defined in v0.1.2). <i>[v0.1.2 · CR-I1-08]</i>
closure_reason	enum	N	0..1	Reason case closed; registry enum closure_reason: OBJECTIVES_MET, INSUFFICIENT_BASIS_TO_CONTINUE, REFERRED, OUT_OF_SCOPE, DUPLICATE, LEGAL_OR_SAFETY_CONSTRAINT, OTHER (minimal set defined in v0.1.2). <i>[v0.1.2 · CR-I1-08]</i>
closed_at	datetime	N	0..1	Time the case entered CLOSED; null otherwise (set by the lifecycle, read-only for clients). The retention trigger CASE_CLOSURE (Section 16.1) uses it. <i>[v0.1.3 · CR-I7-06]</i>

Normative rules:

- Case SHALL contain a documented legitimate purpose before collection begins.
- Case title SHALL avoid presuming criminality.
- Case closure SHALL NOT delete reusable entity or evidence objects.
- A case SHALL NOT be activated (gate transition to ACTIVE) without at least one InvestigationQuestion (SRS-FR-CASE-001). *[v0.1.2 · CR-I1-01]*
- Only the case LEAD (Section 6.3) MAY change the classification or access labels of a Case, and only as an upgrade (higher classification, added labels). Downgrades and label removal SHALL require a recorded reviewer decision; until that workflow exists (increment I6, with review) they SHALL be rejected (403). Every change SHALL generate an AuditEvent. *[v0.1.2 · CR-I1-09]* The reviewer decision is an approved HANDLING_CHANGE review recorded as a HandlingChange (Section 16.4); a direct PATCH downgrade stays 403. *[v0.1.3 · CR-I6-12]*

Lifecycle states:

| DRAFT -> AUTHORIZED -> ACTIVE -> REVIEW -> CLOSED -> MONITORING -> REOPENED

Gate → case-status mapping. The case status changes only through an approved lifecycle gate (Section 6.5) for exactly that transition (database guard): G0 DRAFT → AUTHORIZED; G1 AUTHORIZED → ACTIVE; G4 ACTIVE → REVIEW; G6 ACTIVE or REVIEW → CLOSED (closure_reason from the G6 submission, closed_at set). G2, G3 and G5 change no status. MONITORING and REOPENED remain defined states but are not reachable through the API in v0.1.x; their triggers are deferred to v0.2. *[v0.1.4 · CR-N-03]*

6.2 InvestigationQuestion

Represents a testable analytical question that constrains scope and prevents open-ended surveillance.

Field	Type	Req.	Cardinality	Semantics
case_id	ref	Y	1	Parent case context.
question	text	Y	1	Neutral question.
priority	enum	Y	1	Analytical priority; registry enum investigation_question_priority: PRIMARY, SECONDARY (Methodology §5.2). [v0.1.4 · CR-N-02]
status	enum	Y	1	OPEN, ANSWERED, RETIRED [v0.1.1 · A09]
answer_summary	text	N	0..1	Short evidence-linked answer.

Normative rules:

- Questions SHOULD be framed to permit both incriminating and exculpatory answers.
- Questions are created and retired through the charter (Section 6.4): a question left out of a later charter version becomes RETIRED, never deleted. [v0.1.4 · CR-N-01]

6.3 CaseMembership [v0.1.2 · CR-I1-02]

Explicit, revocable grant of a principal to a case (SRS-FR-SEC-002). Access to case work is defined through case membership.

Field	Type	Req.	Cardinality	Semantics
id	ref	Y	1	Membership identifier.
case	ref	Y	1	Case.
principal	principal	Y	1	Member.
role	enum	Y	1	LEAD, ANALYST, REVIEWER (registry enum case_membership_role).
protected_source_authorized	boolean	Y	1	Explicit per-case grant to see SOURCE_PROTECTED material; effective only together with the protected-source eligibility role (Section 16). Defaults to false.
granted_by	principal	Y	1	Granting principal.
granted_at	datetime	Y	1	Grant time.
revoked_at	datetime	N	0..1	Revocation time (soft revocation).
revoked_by	principal	N	0..1	Revoking principal.

Normative rules:

- A principal SHALL have at most one active (non-revoked) membership per case, and every case SHALL have exactly one active LEAD membership. [v0.1.2 · CR-I1-02]
- Memberships SHALL be managed by the case LEAD. Revocation SHALL be soft (revoked_at), so access history stays reconstructable; the LEAD membership cannot be revoked, the lead is changed through the case. Every grant, change and revocation SHALL generate an AuditEvent. [v0.1.2 · CR-I1-02]
- LEAD is the case owner / investigation lead; REVIEWER and LEAD memberships qualify a principal as reviewer for fact establishment and merge/unmerge (Sections 7.5, 8.4). [v0.1.2 · CR-I1-02, CR-I2-11]

6.4 CharterVersion [v0.1.4 · CR-N-01]

Append-only version of the Investigation Charter of a case (Framework §5, Annex A; Methodology §5.2; SRS-FR-CASE-002). The current version is the one with the highest number; the Case purpose follows it.

Field	Type	Req.	Cardinality	Semantics
case	ref	Y	1	Case.
version	integer	Y	1	1, 2, ... per case.
purpose	text	Y	1	Legitimate investigative purpose.
questions	snapshot[]	Y	1..50	InvestigationQuestions of this version (id, question, priority, status).
change_rationale	text	N*	0..1	REQUIRED for every version after the first (scope change control, Methodology §5.3).
trigger, public_interest_rationale	text	N	0..1	G0 intake fields (Framework §5.2).

Field	Type	Req.	Cardinality	Semantics
scope_included, scope_excluded, initial_subjects, initial_hypotheses, expected_sensitive_data, harm_assessment	text	N	0..1	Scope fields (Framework §5.3).
legal_basis, legal_privacy_constraints, security_source_risks, collection_methods_authorized, collection_methods_prohibited	text	N	0..1	Collection fields (Framework §5.4).
expected_outputs, stop_conditions	text	N	0..1	Annex A.
jurisdictions	code[]	N	0..50	Jurisdictions in scope.
time_period	TemporalValue	N	0..1	Period under investigation.
review_date	date	N	0..1	Next charter review.
created_by, created_at	principal, datetime	Y	1	Author and time.

Normative rules:

- Charter versions SHALL be append-only; an earlier version stays retrievable with actor and time.
- The approved G1 binds to the charter version submitted; a later version flags the charter amended_since_approval (scope change for reviewer judgement, not a new approval requirement in v0.1.4).

6.5 Gate [v0.1.4 · CR-N-03, CR-N-05, CR-N-06, CR-N-07]

One record per lifecycle gate G0-G6 of a case (Framework §5; Methodology §3), with an append-only GateDecision history.

Field	Type	Req.	Cardinality	Semantics
case	ref	Y	1	Case.
gate_code	code	Y	1	G0 ... G6.
gate_status	enum	Y	1	Registry enum gate_status: NOT_SUBMITTED, SUBMITTED, APPROVED, REJECTED.
submitted_by, submitted_at, submit_rationale	principal, datetime, text	N	0..1	Latest submission.
submitted_charter_version	integer	N	0..1	Charter version the submission binds to (G0, G1).
closure_reason	enum	N	0..1	closure_reason; REQUIRED to submit G6, refused for other gates.
decided_by, decided_at, decision_rationale, conditions	principal, datetime, text, text	N	0..1	Latest decision; conditions per CIG CAS-02.
decisions	GateDecision[]	Y	0..n	Append-only: action (SUBMIT, APPROVE, REJECT), actor, time, rationale, conditions, charter version, case status before / after.

Normative rules:

- Status: NOT_SUBMITTED → SUBMITTED → APPROVED | REJECTED; REJECTED → SUBMITTED; APPROVED is final.
- Case-status transitions: G0 DRAFT → AUTHORIZED, G1 AUTHORIZED → ACTIVE, G4 ACTIVE → REVIEW, G6 ACTIVE | REVIEW → CLOSED; G2, G3, G5 none (Section 6.1).
- Readiness criteria the system checks (at submission and again at approval) are listed in API Specification §12; all other criteria of Framework §5 are reviewer judgement recorded in the rationale (Framework §5.9).
- Gates control the case lifecycle only (activation, review, closure); collection, assessment finalisation and dissemination keep their own controls. Per-organisation or per-risk-tier gate configuration is deferred to v0.2.
- The decider SHALL NOT be the submitter (database CHECK) and SHALL hold a LEAD or REVIEWER membership. For a high-impact case (risk_rating HIGH or CRITICAL, or classification RESTRICTED or SOURCE_PROTECTED) the G0 / G1 decider SHALL NOT have authored the charter version under approval.

6.6 Task [v0.1.4 · CR-N-04]

Case-scoped work item (SRS-FR-CASE-004; Screen Inventory SCR-CASE-006).

Field	Type	Req.	Cardinality	Semantics
title	string	Y	1	Short title.
description	text	N	0..1	Details.
task_type	enum	Y	1	Registry enum task_type: COLLECTION, VERIFICATION, ANALYSIS, REVIEW, OTHER.
task_status	enum	Y	1	Registry enum task_status: OPEN, IN_PROGRESS, BLOCKED, DONE, CANCELLED.
assignee	principal	N	0..1	Active, eligible member of the case.
due_date	date	N	0..1	Due date.
depends_on	ref[]	N	0..50	Tasks of the same case.
completed_by,	principal,	N	0..1	Set when DONE.
completed_at	datetime			
history	TaskEvent[]	Y	0..n	Append-only: actor, time, changes, rationale.

Normative rules:

- DONE and CANCELLED are final (database-guarded); DONE requires every dependency to be DONE or CANCELLED.
- Case members with update rights change any field; the assignee may change only the status of their own task.
- Links from tasks to sources, entities and hypotheses are deferred to v0.2.

7. Source, Evidence, Claim, and Fact Model

7.1 Source

Represents the origin or provider of information. A Source describes where information came from; it is not itself the extracted evidentiary proposition.

Field	Type	Req.	Cardinality	Semantics
source_type	enum	Y	1	REGISTRY, COURT_RECORD, MEDIA, WHISTLEBLOWER, WEBSITE, DATASET, DOCUMENT, INTERVIEW, OTHER [v0.1.1 · A09]
publisher_or_origin	string	N	0..1	Originating organization/person.
locator	uri/string	N	0..1	URL, archival location, or repository reference.
accessed_at	datetime	Y	1	Collection/access time.
publication_date	date	N	0..1	When published/issued.
reliability_grade	enum	N	0..1	A-F source reliability.
lawful_access_basis	text/enum	Y	1	Why collection/use is permitted.
archive_ref	ref	N	0..1	Preserved snapshot or evidence item.

Normative rules:

- Source reliability SHALL be separated from credibility of individual information items.
- Anonymous sources SHALL NOT be treated as inherently unreliable; uncertainty SHALL be recorded explicitly.
- Source and EvidenceExtract have no lifecycle; their envelope status is REGISTERED (registry enum envelope_status). [v0.1.2 · CR-I2-03]
- Every case-context object (Source, EvidenceItem, Entity, Relationship, Asset, Ownership-Interest, ControlAssertion, Event, ValueFlow, Indicator, TypologyMatch, Hypothesis, IntelligenceGap, Assessment, IntelligenceProduct) SHALL be created with 1..20 case links; the creator needs update rights on every named case. An object without case context would be reachable by no one. Extracts and derivatives inherit the links of their parent; claims, facts and decisions belong to the case of their path. [v0.1.2 · CR-I2-01, CR-I3-13, CR-I4-12]

7.2 EvidenceItem

Represents a preserved evidentiary object such as a file, page image, registry record, transcript, screenshot, export, or structured record.

Field	Type	Req.	Cardinality	Semantics
source_id	ref	Y	1	Origin source.
evidence_type	enum	Y	1	DOCUMENT, IMAGE, RECORD, TRANSCRIPT, DATASET_ROW, WEB_SNAPSHOT, OTHER [v0.1.1 · A09]
content_hash	string	Y*	0..1	SHA-256 or equivalent where bytes are available.
storage_ref	uri/ref	Y	1	Controlled storage location. Exposed only as an opaque reference (evidence:<id>); bucket and key are never disclosed. [v0.1.2 · CR-I2-06]
acquired_at	datetime	Y	1	Acquisition time.
acquired_by	principal	Y	1	Collector.
original_format	string	N	0..1	MIME/type.
verification_status	enum	Y	1	UNVERIFIED, SOURCE_VERIFIED, INDEPENDENTLY_CORROBORATED, DISPUTED [v0.1.1 · A09]
redaction_state	enum	Y	1	NONE, WORKING_REDACTION, PUBLICATION_REDACTION [v0.1.1 · A09]
byte_size	integer	N	0..1	Size of the original bytes. [v0.1.2 · CR-I2-06]
hash_algorithm	string	N	0..1	Algorithm of content_hash (e.g. SHA-256). [v0.1.2 · CR-I2-06]
derived_from	ref	N	0..1	Parent EvidenceItem of a derivative. [v0.1.2 · CR-I2-06]
derivation_type	text	N	0..1	Kind of derivative (redaction, OCR text, ...); free text in v0.1.2. [v0.1.2 · CR-I2-06] Registered open vocabulary derivation_type: OCR_TEXT, TEXT_EXTRACTION, MACHINE_TRANSLATION (produced only by the derived-text worker, Section 7.7); other UPPER_SNAKE_CASE values (e.g. REDACTION) are accepted for registered derivatives. [v0.1.4 · CR-N-09]

Normative rules:

- Original evidence bytes SHALL be immutable after registration except for controlled preservation migrations.
- Derived/redacted copies SHALL be linked to the original by derivation relationship.

Lifecycle states:

| COLLECTED -> REGISTERED -> VERIFIED -> SUPERSEDED -> RESTRICTED

7.3 EvidenceExtract

Represents the precise portion of EvidenceItem used analytically.

Field	Type	Req.	Cardinality	Semantics
evidence_id	ref	Y	1	Parent evidence.
locator	string	Y	1	Page, paragraph, timecode, row, bounding region, or equivalent.
extract_text	text	N	0..1	Verbatim/normalized extract when lawful.
extract_hash	string	N	0..1	Integrity value for extracted content.
analyst_note	text	N	0..1	Context note; SHALL remain distinguishable from extract.

7.4 Claim

Represents a proposition asserted by a source or person. A Claim is not automatically accepted as fact. A Claim is a permanent record of what a source asserts; it is never converted into, upgraded to, or overwritten by a Fact. [v0.1.1 · A10]

Field	Type	Req.	Cardinality	Semantics
subject_refs	ref[]	Y	1..n	Entities/events referenced.
predicate	string	Y	1	Controlled or human-readable predicate.
object_value	variant	Y	1	Claimed value/entity.
evidence_extract_refs	ref[]	Y	1..n	Supporting extracts.
claimant	ref/string	N	0..1	Who makes the assertion.

Field	Type	Req.	Cardinality	Semantics
credibility_grade	enum	N	0..1	1-6 information credibility; registry enum credibility_grade (INDEPENDENTLY_CONFIRMED, PROBABLY_TRUE, POSSIBLY_TRUE, DOUBTFUL, IMPROBABLE, CANNOT_BE_JUDGED) whose wire values are the digit codes "1"..."6" (registry exception). [v0.1.2 · CR-12-08]
claim_status	enum	Y	1	RECORDED, UNDER_REVIEW, CORROBORATED, CONTRADICTED, UNRESOLVED [v0.1.1 · A10]
disputed	boolean	Y	1	Whether contested. Retained for compatibility; derived as true when claim_status = CONTRADICTED or an open dispute exists. [v0.1.1 · A10]

Normative rules:

- Claims SHALL preserve attribution.
- A Fact SHALL NOT be created on the basis of a source assertion solely because it appears in an official document if the document merely records a third-party allegation. [v0.1.1 · A10]
- The recorded assertion of a Claim (subject, predicate, object value, claimant, evidence extracts) SHALL NOT be overwritten by analyst conclusions. Analytical outcomes are recorded as VerificationDecision objects (Section 7.6) and reflected in claim_status. [v0.1.1 · A10]

Lifecycle states: [v0.1.1 · A10]

| RECORDED -> UNDER_REVIEW -> CORROBORATED | CONTRADICTED | UNRESOLVED

Every transition out of RECORDED SHALL be backed by a VerificationDecision. [v0.1.1 · A10]

A decided claim (CORROBORATED, CONTRADICTED or UNRESOLVED) MAY return to UNDER_REVIEW through a further VerificationDecision; earlier decisions are kept. Any other transition SHALL be rejected (409). Claim decisions MAY be recorded by members with role LEAD, ANALYST or REVIEWER (Section 6.3). [v0.1.2 · CR-12-11]

7.5 Fact

Represents a proposition accepted by the investigation as established to the stated confidence threshold. A Fact is a separate analytical object supported by one or more evidence items (mandatory), optionally by one or more Claims, and by one or more VerificationDecisions; it is not produced by transforming a Claim. [v0.1.1 · A10] [v0.1.1 · C02]

Field	Type	Req.	Cardinality	Semantics
proposition	text/ structured	Y	1	Established proposition.
supporting_evidence	ref[]	Y	1..n	Evidence supporting acceptance; EvidenceItem or EvidenceExtract only, readable by the creator and in the case context (422 otherwise). [v0.1.2 · CR-12-07]
contradicting_evidence	ref[]	N	0..n	Known contradictory material.
supporting_claim_refs	ref[]	N	0..n	Claims that support the fact, if any. The Claims remain unchanged. [v0.1.1 · A10]
fact_status	enum	Y	1	PROVISIONAL, ESTABLISHED, DISPUTED, SUPERSEDED [v0.1.1 · A09, A10]
verification_decision_refs	ref[]	Y	1..n	VerificationDecision records that created and revised the fact (append-only history). Server-populated and read-only: the CREATE decision is written atomically with the Fact, and each later fact command appends its decision. [v0.1.1 · A10] [v0.1.1 · C01]
proposed_by	principal	Y	1	Principal who proposed the fact. [v0.1.1 · A10]
superseded_by	ref	Y*	0..1	Replacement Fact; required when fact_status = SUPERSEDED. [v0.1.1 · A10]
valid_time	interval	N	0..1	When proposition is true in real world.
confidence	confidence	Y	1	Analytical confidence.

Normative rules:

- Fact status SHALL be revisable when materially new evidence emerges.

- Fact objects SHALL NOT encode legal guilt or criminal liability unless directly quoting an authoritative adjudication, in which case attribution SHALL be explicit.

Claim and Fact lifecycle rules [v0.1.1 · A10]

Approved by product owner, 2026-10-08. The rules below and Section 7.6 were added in v0.1.1 to close audit finding A10. [v0.1.1 · A10]

- Fact creation.** A Fact SHALL be created only with (a) one or more evidence references (supporting_evidence, mandatory, 1..n), optionally one or more supporting claim references (supporting_claim_refs, 0..n), and (b) a decision rationale. The system SHALL create the Fact and its CREATE VerificationDecision atomically in the same transaction; the CREATE decision targets the new Fact, and neither record SHALL exist without the other. [v0.1.1 · C01] [v0.1.1 · C02] A newly created Fact SHALL start as PROVISIONAL. Creating a Fact SHALL NOT modify, convert, or close the supporting Claims. A Claim SHALL NOT be treated as a Fact without such a decision. [v0.1.1 · A10]
- Who may act.** An Investigator/Analyst MAY record Claims and propose PROVISIONAL Facts. Moving a Fact to ESTABLISHED SHALL require a Reviewer who is not the proposer; a Reviewer is a principal with REVIEWER or LEAD membership on the fact's case (403 otherwise, and always 403 for the proposer). [v0.1.2 · CR-I2-11] Any authorized case member MAY move a Fact to DISPUTED with supporting evidence. Moving a Fact to SUPERSEDED SHALL require a reference to the replacement Fact (superseded_by). [v0.1.1 · A10]
- Revision.** Every status change SHALL be recorded as a new VerificationDecision; earlier decisions and prior states SHALL be preserved. [v0.1.1 · A10] The ESTABLISH, DISPUTE and SUPERSEDE decisions SHALL be created atomically with the corresponding status change by the fact command itself. [v0.1.1 · C01]
- Dependent flagging.** When a Fact becomes DISPUTED or SUPERSEDED, every dependent Assessment and IntelligenceProduct SHALL be flagged review_required with a link to the triggering VerificationDecision (review_trigger_ref). Published or disseminated products SHALL NOT be mutated; a correction review task SHALL be created instead. History SHALL be preserved. [v0.1.1 · A10]
- Indirect dependents.** Flagging SHALL cover direct dependents (the Fact in supporting_refs) and indirect ones: Indicators whose evidence includes the Fact, TypologyMatches citing such Indicators, and Hypotheses with a SUPPORTS matrix cell on such an object, together with the Assessments and products depending on them. Drafts and finalized objects are flagged alike; the recorded judgement is not changed. A flagged draft SHALL NOT be finalized (409 REVIEW_REQUIRED); clearing review_required is a review action (I6). The dependents listing SHALL show only dependents the caller may read. [v0.1.2 · CR-I4-11]

Fact lifecycle: [v0.1.1 · A10]

```
PROVISIONAL -> ESTABLISHED
PROVISIONAL | ESTABLISHED -> DISPUTED
PROVISIONAL | ESTABLISHED | DISPUTED -> SUPERSEDED (superseded_by required)
```

7.6 VerificationDecision [v0.1.1 · A10]

Represents a single, append-only verification outcome on a Claim or Fact (for example: claim corroborated or contradicted; fact created, established, disputed, or superseded). [v0.1.1 · A10]

Field	Type	Req.	Cardinality	Semantics
target_ref	ref	Y	1	Claim or Fact the decision applies to. Outcome; for Claims a claim_status value (UNDER_REVIEW, CORROBORATED, CONTRADICTED, UNRESOLVED); for Facts a fact action (CREATE, ESTABLISH, DISPUTE, SUPERSEDE). CREATE (display label "Create fact") records the decision that creates a Fact supported by the referenced evidence (and claims, if any); it does not convert or alter any Claim. [v0.1.1 · A10] For Facts, decisions are never created on their own: CREATE is written atomically with the new Fact, and ESTABLISH/DISPUTE/SUPERSEDE atomically with the fact command. [v0.1.1 · C01]
decision	enum	Y	1	
rationale	text	Y	1	Reasoning for the decision.

Field	Type	Req.	Cardinality	Semantics
evidence_refs	ref[]	Y	1..n	Evidence or extracts relied on; EvidenceItem or EvidenceExtract only, readable and in the case context (422 otherwise). [v0.1.2 · CR-I2-07]
decided_by	principal	Y	1	Decision maker.
decided_at	datetime	Y	1	Decision time.
review_ref	ref	N	0..1	Related Review, if any.

Normative rules:

- VerificationDecision records SHALL be append-only; a mistaken decision is corrected by a later decision, not by editing or deleting the earlier one.
- A decision moving a Fact to ESTABLISHED SHALL have decided_by different from the Fact's proposed_by.
- Each VerificationDecision SHALL generate an AuditEvent.

7.7 DerivedText [v0.1.4 · CR-N-09, CR-N-10]

Record of one automated transformation of an evidence item's content into text: OCR, text extraction or offline machine translation (Product and Feature Specification F-DOC-002 "DerivedArtifact"; SRS-FR-EVD-005, SRS-FR-DOC-002, SRS-FR-DOC-003, SRS-AI-001). A case-context object; classification, access labels and case links are inherited from its input (DM-I14). It is never the original evidence and never a fact.

Field	Type	Req.	Cardinality	Semantics
transformation	enum	Y	1	Registry enum derived_text_transformation: OCR, TEXT_EXTRACTION, MACHINE_TRANSLATION.
status	enum	Y	1	Registry enum derived_text_status: QUEUED, PROCESSING, REVIEW_REQUIRED, REVIEWED, FAILED.
source_evidence	ref	Y	1	EvidenceItem whose bytes were read (an original or a derivative).
root_evidence	ref	Y	1	Original at the root of the lineage.
input_text	ref	N	0..1	DerivedText translated (translations of an OCR / extraction output).
evidence	ref	N	0..1	Derivative EvidenceItem holding the output document (derivation_type OCR_TEXT, TEXT_EXTRACTION or MACHINE_TRANSLATION).
automated	boolean	Y	1	Always true.
label	string	Y	1	E.g. "OCR text — machine-generated, review required", "Machine translation — review required".
engine,	string	N	0..1	Offline engine and version.
engine_version	string	N	0..1	Offline engine and version.
models	object[]	N	0..n	Name, version, SHA-256 and licence of each model / language file.
languages;	code[]; code	N	0..n	OCR languages; translation pair.
source_language,	code[]; code	N	0..n	OCR languages; translation pair.
target_language	code[]; code	N	0..n	OCR languages; translation pair.
page_count,	integer, number	N	0..1	Mean OCR word confidence (not an analytical confidence).
mean_confidence	integer, number	N	0..1	Mean OCR word confidence (not an analytical confidence).
input_sha256,	string	N	0..1	Hash of the input read and of the output.
content_sha256	string	N	0..1	Hash of the input read and of the output.
pages	DerivedTextPage[]	N	0..n	page_number, source_page_number (page mapping), method (registry enum text_page_method: OCR, TEXT_LAYER, PLAIN_TEXT, MACHINE_TRANSLATION), confidence, text, source_text (translations).
error_code	string	N	0..1	Safe code when FAILED.
requested_by,	principal, principal,	N	0..1	Requester and the recorded human review.
reviewed_by,	datetime, text	N	0..1	Requester and the recorded human review.
reviewed_at,	datetime, text	N	0..1	Requester and the recorded human review.
review_note	text	N	0..1	Requester and the recorded human review.

Normative rules:

- The original SHALL only be read; it is re-hashed before processing and a mismatch fails the record.
- Status moves forward only; a completed or failed record's output, engine, lineage and hashes are immutable; pages are append-only; a recorded review is immutable (database guards).

- Machine output is REVIEW_REQUIRED until a person reviews it. Review does not create facts; a Fact still needs its VerificationDecision against evidence (CIG TEC-01).
- Translations keep the source-language text of every page, are never translated again, and keep the machine-translation label in any intelligence product that uses them (Methodology §28.1).
- Every automation engine runs offline; evidence content never leaves the host (CIG EVD-02, TEC-01).

8. Entity and Identity Model

8.1 Entity

Canonical identity-bearing node used across cases.

Field	Type	Req.	Cardinality	Semantics
entity_type	enum	Y	1	PERSON, ORGANIZATION, ACCOUNT, ADDRESS, DOMAIN, PHONE, EMAIL, WALLET, PROPERTY, VEHICLE, VESSEL, AIRCRAFT, CONTRACT, PROJECT, OTHER [v0.1.1 · A09]
primary_name	string	Y	1	Preferred display label.
aliases	string[]	N	0..n	Alternative labels.
identifiers	identifier[]	N	0..n	Typed external identifiers.
resolution_status	enum	Y	1	UNRESOLVED, RESOLVED, CONFLICTED, MERGED, SPLIT [v0.1.1 · A09]
resolution_confidence	confidence	N	0..1	Identity match confidence.
canonical_parent	ref	N	0..1	Target if merged; set from the surviving_entity_ref of the MERGE ResolutionDecision (Section 8.4). [v0.1.1 · ER]
source_refs	ref[]	Y	1..n	Sources establishing identity.

Normative rules:

- Names SHALL NOT be treated as unique identifiers.
- A merged entity SHALL retain links to precursor records and merge rationale.
- Implementations SHALL support reversing a mistaken merge without losing history.
- resolution_status is entity state. It SHALL change only as the effect of a ResolutionDecision (Section 8.4); decisions about pairs or sets of records are not stored in resolution_status. [v0.1.1 · ER]

8.2 PersonProfile

Extension of Entity for natural persons. Sensitive attributes SHALL only be stored when relevant and lawful.

Field	Type	Req.	Cardinality	Semantics
entity_id	ref	Y	1	Base Entity.
birth_date	date/partial	N	0..1	Known/partial DOB.
nationalities	code[]	N	0..n	Known nationalities.
occupation	string[]	N	0..n	Relevant occupation/role.
public_official_status	enum	N	0..1	NONE, CURRENT, FORMER, UNKNOWN; not equivalent to suspicion. [v0.1.1 · A09]
sensitive_attribute_notes	restricted text	N	0..1	Only if strictly necessary and lawful.

Normative rules:

- Protected characteristics SHALL NOT be used as AML indicators by themselves.

8.3 OrganizationProfile

Extension for companies, NGOs, agencies, trusts, partnerships, and other organized bodies.

Field	Type	Req.	Cardinality	Semantics
entity_id	ref	Y	1	Base Entity.
organization_type	enum	Y	1	COMPANY, NGO, GOVERNMENT, TRUST, PARTNERSHIP, ASSOCIATION, OTHER [v0.1.1 · A09]
registration_number	string	N	0..1	Legal registration identifier.
registration_jurisdiction	code	N	0..1	Jurisdiction.
incorporation_date	date	N	0..1	Formation date.
dissolution_date	date	N	0..1	If dissolved.
registered_address	ref	N	0..1	Address entity.

8.4 ResolutionDecision [v0.1.1 · ER]

Append-only record of a decision about whether two or more Entity records refer to the same real-world subject. Separates decisions about pairs/sets of records from the state of each Entity. Approved by product owner, 2026-10-08.

Field	Type	Req.	Cardinality	Semantics
subject_entity_refs	ref[]	Y	2..n	Entity records compared.
decision	enum	Y	1	MERGE, KEEP_SEPARATE, POSSIBLE_MATCH, DEFER, UNMERGE
surviving_entity_ref	ref	C	0..1	Required for MERGE.
reverses_decision_ref	ref	C	0..1	Required for UNMERGE (the MERGE decision being reversed).
matching_attributes	structured[]	N	0..n	Attributes that agree.
conflicting_attributes	structured[]	N	0..n	Attributes that disagree.
evidence_refs	ref[]	Y	1..n	Evidence considered; EvidenceExtract, EvidenceItem or Source, readable and in the case context (422 otherwise). [v0.1.2 · CR-I2-07]
confidence	confidence	Y	1	Confidence object (HIGH, MODERATE, LOW, INSUFFICIENT_BASIS + basis; Section 14.1).
rationale	text	Y	1	Why the decision was made.
decided_by	principal	Y	1	Decision maker.
decided_at	datetime	Y	1	Decision time.
reviewer_ref	principal	C	0..1	Required for every MERGE and UNMERGE (all are high-impact in the MVP); a principal with REVIEWER or LEAD membership on the case of every subject entity (422 otherwise); SHALL differ from decided_by (409 STATE_CONFLICT). [v0.1.2 · CR-I2-10, CR-I2-09]

Normative rules:

- ResolutionDecision records SHALL be append-only; a mistaken decision is corrected by a later decision (for example UNMERGE reversing a MERGE), not by editing or deleting the earlier one.
- Effects on Entity state: MERGE → each absorbed record becomes MERGED with canonical_parent = surviving_entity_ref, and the survivor becomes RESOLVED; UNMERGE → restored records become SPLIT; nothing is re-attributed (see the logical-repointing rule below); POSSIBLE_MATCH → a candidate link is recorded with no state change; KEEP_SEPARATE → no state change, and the same pair SHALL NOT be re-suggested unless new evidence is attached; DEFER → state stays or becomes UNRESOLVED.
- Each ResolutionDecision SHALL generate an AuditEvent (action MERGE for MERGE, SPLIT for UNMERGE, otherwise UPDATE).
- Merge is logical repointing (canonical model): absorbed records point to the survivor through canonical_parent, and relationships, claims and absorbed records are never rewritten. UNMERGE therefore needs no re-attribution: it removes what the merge added to the survivor and restores the survivor's prior handling. [v0.1.2 · CR-I2-13]
- Every MERGE and UNMERGE is high-impact in the MVP and requires a reviewer_ref distinct from the decider. [v0.1.2 · CR-I2-10]

9. Relationship, Ownership, and Control Model

9.1 Relationship

Represents a typed, evidence-linked edge between entities. Relationships are first-class objects because their source, time, and confidence may differ.

Field	Type	Req.	Cardinality	Semantics
from_entity	ref	Y	1	Origin node.
relationship_type	enum	Y	1	Typed semantic relationship.
to_entity	ref	Y	1	Target node: an Entity, or an Asset (the response states to_object_type). [v0.1.2 · CR-I3-08]
directionality	enum	Y	1	DIRECTED, SYMMETRIC [v0.1.1 · A09]
valid_from	date/datetime	N	0..1	Relationship start.
valid_to	date/datetime	N	0..1	Relationship end.
supporting_evidence	ref[]	Y	1..n	Evidence: EvidenceExtract, EvidenceItem or Fact, readable and in the case context (422 otherwise). [v0.1.2 · CR-I2-07]
confidence	confidence	Y	1	Confidence; confidence.basis carries the basis of the relationship (DM-I06). [v0.1.2 · CR-I2-15]
relationship_status	enum	Y	1	ASSERTED, ESTABLISHED, DISPUTED, SUPERSEDED [v0.1.1 · A09]

Normative rules:

- Edges in a graph visualization SHALL retain their evidence links in the underlying model.
- A relationship based only on co-occurrence SHALL NOT be mislabeled as control, ownership, or financial transfer.
- A Relationship created directly MAY target an Asset only with type USES, MANAGES or ACQUIRED. OWNS, BENEFICIAL_OWNER_OF and CONTROLS to an Asset SHALL be created only together with an OwnershipInterest or ControlAssertion (422 otherwise). [v0.1.2 · CR-I3-08]

9.2 OwnershipInterest

Specialized relationship for legal or beneficial ownership.

Field	Type	Req.	Cardinality	Semantics
owner_entity	ref	Y	1	Owner/beneficial owner.
owned_entity_or_asset	ref	Y	1	Company/share/asset.
ownership_type	enum	Y	1	LEGAL, BENEFICIAL, ECONOMIC_INTEREST, NOMINEE_ASSERTED [v0.1.1 · A09]
percentage	decimal	N	0..1	0-100 if known.
basis	text/enum	Y	1	Registry, filing, court record, reconstruction, etc.
supporting_evidence	ref[]	Y	1..n	Evidence.
confidence	confidence	Y	1	Confidence.

Normative rules:

- Legal ownership and beneficial ownership SHALL be stored as distinct types.
- Nominee or beneficial ownership inferred from public-source patterns SHALL be marked as inferred/asserted, not established, unless evidence supports establishment.
- An OwnershipInterest of type BENEFICIAL, ECONOMIC_INTEREST or NOMINEE_ASSERTED SHALL become ESTABLISHED only when its supporting_evidence includes an ESTABLISHED Fact (422 otherwise). [v0.1.2 · CR-I3-11]
- OwnershipInterest (interest_status) and ControlAssertion (assertion_status) use the relationship_status values ASSERTED, ESTABLISHED, DISPUTED, SUPERSEDED; their envelope status mirrors that value. Supporting evidence: EvidenceExtract, EvidenceItem or Fact. A ControlAssertion's controlled_entity MAY be an Asset. [v0.1.2 · CR-I3-05, CR-I3-01]

9.3 ControlAssertion

Represents control that may exist without formal ownership.

Field	Type	Req.	Cardinality	Semantics
controller	ref	Y	1	Person/entity exercising control.
controlled_entity	ref	Y	1	Target.
control_basis	enum[]	Y	1..n	VOTING, APPOINTMENT, FINANCING, CONTRACTUAL, OPERATIONAL, FAMILY_PROXY, OTHER [v0.1.1 · A09]
control_level	enum	N	0..1	MINOR, SIGNIFICANT, DOMINANT, UNKNOWN [v0.1.1 · A09]
supporting_evidence	ref[]	Y	1..n	Evidence.
confidence	confidence	Y	1	Confidence.

10. Asset and Event Model

10.1 Asset

Represents an item or right with economic value.

Field	Type	Req.	Cardinality	Semantics
asset_type	enum	Y	1	PROPERTY, VEHICLE, VESSEL, AIRCRAFT, SECURITY, COMPANY_SHARE, CRYPTO_ASSET, PRECIOUS_METAL, LUXURY_GOOD, INTELLECTUAL_PROPERTY, OTHER [v0.1.1 · A09]
description	text	Y	1	Neutral description.
legal_owner	ref	N	0..1	Known legal owner.
beneficial_owner_assertions	ref[]	N	0..n	OwnershipInterest refs.
controller_refs	ref[]	N	0..n	Control assertions.
valuation	money/ range	N	0..1	Value or range.
valuation_date	date	N	0..1	Valuation date.
valuation_basis	text	C	0..1	Registry, market estimate, appraisal, reported value. Free text in v0.1.2 (vocabulary deferred); required when a valuation is given. [v0.1.2 · CR-I3-06]
location_ref	ref	N	0..1	Location entity.

Normative rules:

- Observed use or association SHALL NOT be represented as ownership without an ownership basis.
- Asset has no lifecycle; its envelope status is REGISTERED. Asset evidence: Source, EvidenceItem or EvidenceExtract. [v0.1.2 · CR-I3-05, CR-I3-01]

10.2 Event

Represents a temporally bounded occurrence involving one or more entities.

Field	Type	Req.	Cardinality	Semantics
event_type	enum	Y	1	INCORPORATION, APPOINTMENT, RESIGNATION, CONTRACT_AWARD, ACQUISITION, DISPOSAL, TRANSFER, COURT_EVENT, PAYMENT_EVENT, PUBLICATION, OTHER [v0.1.1 · A09]
description	text	N	0..1	Neutral description. [v0.1.2 · CR-I3-02]
start_time	datetime/date	Y*	0..1	Start/occurrence; null only when time_precision = UNKNOWN (unknown dates, Methodology §14.2). [v0.1.2 · CR-I3-06]
time_precision	enum	Y	1	DATETIME, DAY, MONTH, YEAR, RANGE, UNKNOWN (registry enum temporal_precision); the string shape SHALL match the precision. [v0.1.2 · CR-I3-06]
approximate	boolean	N	0..1	Marks an approximate date. [v0.1.2 · CR-I3-02]
end_time	datetime/date	N	0..1	End if interval.
participant_refs	ref[]	Y	1..n	Entities participating.
asset_refs	ref[]	N	0..n	Assets involved. [v0.1.2 · CR-I3-02]
location_ref	ref	N	0..1	Location.
evidence_refs	ref[]	Y	1..n	Supporting evidence: EvidenceExtract, EvidenceItem or Fact. [v0.1.2 · CR-I3-10]
confidence	confidence	Y	1	Confidence.

Normative rules:

- Event has no stored lifecycle in v0.1.2; its envelope status is RECORDED. The Methodology §14.1 event statuses (confirmed / probable / possible / disputed) are not registered and are deferred to v0.2. [v0.1.2 · CR-I3-05]

11. Value-Flow Model

Mandatory distinction

Every ValueFlow SHALL carry an epistemic class: DIRECT, DOCUMENTED, RECONSTRUCTED, or HYPOTHETICAL. Systems SHALL preserve this distinction in storage, APIs, exports, and visualizations.

11.1 ValueFlow

Represents movement, conversion, allocation, or inferred transfer of economic value.

Field	Type	Req.	Cardinality	Semantics
flow_class	enum	Y	1	DIRECT, DOCUMENTED, RECONSTRUCTED, HYPOTHETICAL [v0.1.1 · A09]
flow_type	enum	Y	1	PAYMENT, CONTRACT, SUBCONTRACT, LOAN, INVESTMENT, ASSET_PURCHASE, ASSET_SALE, GRANT, DONATION, DIVIDEND, CRYPTO_TRANSFER, VALUE_CONVERSION, OTHER [v0.1.1 · A09]
origin_ref	ref	Y	1	Origin entity/asset/event.
destination_ref	ref	Y	1	Destination entity/asset/event.
amount	money/range	N	0..1	Known/estimated amount.
currency	code	N	0..1	Currency if monetary.
occurred_at	datetime/date/interval	N	0..1	Temporal scope.
evidence_refs	ref[]	Y*	0..n	Required for direct/documentated/reconstructed.
reconstruction_basis	text	Y*	0..1	Required for reconstructed/hypothetical.
confidence	confidence	Y	1	Confidence.
disclaimer	text	N	0..1	Required presentation caveat if needed.

Normative rules:

- DIRECT flows require evidence of the actual movement of value.
- DOCUMENTED flows require an authoritative or transactional document evidencing a transfer/obligation but not necessarily settlement.
- RECONSTRUCTED flows SHALL identify the inferential steps linking origin and destination.
- HYPOTHETICAL flows SHALL never be included in an external intelligence product as if observed.
- Evidence references of flows, legs and events SHALL be EvidenceExtract, EvidenceItem or Fact, readable and in the case context. DIRECT and DOCUMENTED flows and legs SHALL reference at least one EvidenceItem or EvidenceExtract; a Fact alone is insufficient (422). [v0.1.2 · CR-I3-10]
- **Certainty order (invariants only).** flow_class is an epistemic category and the registry stays unordered for display. For invariants only, the order HYPOTHETICAL < RECONSTRUCTED < DOCUMENTED < DIRECT applies; it SHALL NOT be displayed, averaged or summed. A change that raises a flow's class SHALL add at least one evidence reference not previously attached; a flow SHALL never be stronger than its weakest leg; lowering a class SHALL meet the target class's requirements (e.g. reconstruction_basis). [v0.1.2 · CR-I3-09]
- ValueFlow has no stored lifecycle; its envelope status is RECORDED. [v0.1.2 · CR-I3-05]
- **Aggregation.** Value SHALL be aggregated per (flow_class, flow_type, currency) with no grand total across classes; CONTRACT and SUBCONTRACT groups are obligations, not settlements, and only DIRECT groups count as settlement evidenced; lower/upper/exact totals

SHALL count unknown amounts and never treat them as 0; legs SHALL NOT be added to their flow's totals. Details: API Specification v0.1.4 §15. *[v0.1.2 · CR-I3-12]*

11.2 ValueFlowLeg

Optional decomposition of a complex flow into ordered legs.

Field	Type	Req.	Cardinality	Semantics
value_flow_id	ref	Y	1	Parent flow.
sequence	integer	Y	1	Order.
origin_ref	ref	Y	1	Leg origin.
destination_ref	ref	Y	1	Leg destination.
amount	money/range	N	0..1	Leg amount.
flow_class	enum	Y	1	Epistemic class for this leg.
evidence_refs	ref[]	N	0..n	Evidence for leg.
reconstruction_basis	text	Y*	0..1	Required for RECONSTRUCTED and HYPOTHETICAL legs. <i>[v0.1.2 · CR-I3-04]</i>
confidence	confidence	N	0..1	Per-leg confidence (SRS-FR-VAL-002). <i>[v0.1.2 · CR-I3-04]</i>

Normative rules:

- Each leg SHALL keep its own evidence, class, value and confidence; the per-class requirements of Section 11.1 apply per leg. *[v0.1.2 · CR-I3-04]*
- Legs SHALL be immutable and append-only (no update or delete): appended in order (sequence = n + 1), each starting where the previous leg ended; a chain that reached the flow's destination takes no more legs (409), and the endpoints of a chained flow are fixed (409). Leg inputs SHALL NOT exceed the flow's classification. *[v0.1.2 · CR-I3-04]*

12. Indicator and Typology Model

12.1 Indicator

Represents an observed condition relevant to analysis. It SHALL NOT be treated as proof of wrongdoing.

Field	Type	Req.	Cardinality	Semantics
indicator_code	string	Y	1	Stable catalogue or local code. A catalogue indicator is referenced by its catalogue indicator ID (<typology_id>-I<nn>, plus catalogue version) and takes its class from the catalogue (a contradicting class → 422); a local indicator uses a LOCAL-... code and an explicit class. <i>[v0.1.2 · CR-I4-02]</i>
indicator_class	enum	Y	1	MECHANISM, CORROBORATING, CONTEXTUAL, DISCONFIRMING, GAP <i>[v0.1.1 · A09]</i>
description	text	Y	1	Observed condition.
subject_refs	ref[]	Y	1..n	Affected objects: Entity, Asset, Event, ValueFlow or Relationship. <i>[v0.1.2 · CR-I4-02]</i>
evidence_refs	ref[]	Y	1..n	Evidence: EvidenceExtract, EvidenceItem or Fact. <i>[v0.1.2 · CR-I4-02]</i>
status	enum	Y	1	OBSERVED, CORROBORATED, DISPUTED, RETIRED <i>[v0.1.1 · A09]</i>
confidence	confidence	Y	1	Confidence.

Normative rules:

- Indicator status transitions: OBSERVED → CORROBORATED / DISPUTED / RETIRED; DISPUTED → OBSERVED / CORROBORATED / RETIRED; RETIRED is terminal. Every change SHALL carry a rationale. Indicators are never deleted and are never proof (responses state is_proof = false). *[v0.1.2 · CR-I4-02]*

12.2 TypologyMatch

Represents analytical consistency between case evidence and a catalogue typology.

Field	Type	Req.	Cardinality	Semantics
typology_id	string/ref	Y	1	Catalogue entry.
subject_refs	ref[]	Y	1..n	Entities/flows under analysis.
indicator_refs	ref[]	Y	1..n	Observed indicators.
disconfirming_refs	ref[]	N	0..n	Contrary evidence/indicators.
consistency_level	enum	Y	1	NO_BASIS, WEAK, PLAUSIBLE, STRONG, COMPELLING [v0.1.1 · A09]
analyst_rationale	text	Y	1	Why level was assigned.
confidence	confidence	Y	1	Confidence.

Normative rules:

- TypologyMatch SHALL express consistency, not guilt or legal classification.
- Strong or compelling consistency SHOULD require multiple independent indicators or direct authoritative evidence, consistent with the Typology Catalogue.
- The analyst assigns consistency_level; the system computes a ceiling (computed_ceiling, with a rule_trace) from the cited indicators, their sources and direct authoritative evidence, and SHALL reject a level above it (422); the system never raises a level. The thresholds are those of Typology Catalogue v0.1.2 §4.1, adopted PROVISIONALLY and subject to AML-specialist review. A match also records indicator_assessments (catalogue observation status per catalogue indicator) and direct_evidence_refs; STRONG and COMPELLING set reviewer_required; a cited indicator becoming DISPUTED or RETIRED flags the match review_required. Envelope status: RECORDED. [v0.1.2 · CR-I4-03, CR-I3-05]

13. Hypothesis, Gap, and Assessment Model

13.1 Hypothesis

Represents a testable analytical explanation.

Field	Type	Req.	Cardinality	Semantics
statement	text	Y	1	Specific falsifiable hypothesis.
hypothesis_type	enum	N	0..1	OWNERSHIP, CONTROL, VALUE_FLOW, TYPOLOGY, SOURCE_OF_FUNDS, OTHER [v0.1.1 · A09]
supporting_refs	ref[]	N	0..n	Evidence/facts/indicators supporting.
contradicting_refs	ref[]	N	0..n	Evidence/facts contradicting.
alternative_hypothesis_refs	ref[]	N	0..n	Competing explanations.
status	enum	Y	1	OPEN, SUPPORTED, WEAKENED, REJECTED, INCONCLUSIVE [v0.1.1 · A09]
confidence	confidence	Y	1	Current confidence.
next_test	text	N	0..1	Most useful discriminating test.
role	enum	N	0..1	PRINCIPAL, ALTERNATIVE_LEGITIMATE, ALTERNATIVE_MECHANISM, INSUFFICIENT_INFORMATION (registry enum hypothesis_role; Methodology §18.1). [v0.1.2 · CR-I4-06]
assumptions	text[]	N	0..n	Stated assumptions. [v0.1.2 · CR-I4-06]

Normative rules:

- A hypothesis SHALL remain open to disconfirmation.
- Rejected hypotheses SHOULD remain auditable rather than deleted.
- **HypothesisLink (matrix cell).** The support/contradiction matrix SHALL be stored as append-only HypothesisLink records: hypothesis, target_ref, effect (SUPPORTS, CONTRADICTS, NEUTRAL, UNKNOWN; registry enum hypothesis_link_effect), rationale, recorded_by, recorded_at. The current cell per target is the latest record; history is kept. supporting_refs and contradicting_refs are derived from the current cells; a removed reference becomes NEUTRAL. [v0.1.2 · CR-I4-04, CR-I4-07]
- A status change SHALL carry a rationale; SUPPORTED requires a SUPPORTS cell and WEAKENED or REJECTED a CONTRADICTS cell (409 otherwise). Status changes are kept as history. [v0.1.2 · CR-I4-08, CR-I4-06]

13.2 IntelligenceGap

Represents a material unknown that limits assessment.

Field	Type	Req.	Cardinality	Semantics
description	text	Y	1	Unknown or missing information.
importance	enum	Y	1	LOW, MEDIUM, HIGH, CRITICAL [v0.1.1 · A09]
related_refs	ref[]	Y	1..n	Hypotheses/questions affected.
collection_feasibility	enum	Y	1	AVAILABLE, DIFFICULT, UNAVAILABLE, UNLAWFUL, OUT_OF_SCOPE [v0.1.1 · A09]
status	enum	Y	1	OPEN, PARTIALLY_RESOLVED, RESOLVED, ACCEPTED [v0.1.1 · A09]
closure_rationale	text	C	0..1	Required for PARTIALLY_RESOLVED, RESOLVED and ACCEPTED. [v0.1.2 · CR-I4-05]

Normative rules:

- IntelligenceGaps SHALL never be deleted; status changes are kept as history. [v0.1.2 · CR-I4-05]

13.3 Assessment

Represents a reasoned analytical judgment supported by evidence and explicit confidence.

Field	Type	Req.	Cardinality	Semantics
judgment	text	Y	1	Analytical conclusion.
scope_refs	ref[]	Y	1..n	Question/hypothesis/subjects addressed: Hypothesis, TypologyMatch, Indicator, Entity, Asset, Event, ValueFlow or Relationship (InvestigationQuestion once available). [v0.1.2 · CR-I4-14]
supporting_refs	ref[]	Y	1..n	Facts/indicators/evidence: Fact, Indicator, TypologyMatch, Hypothesis, EvidenceExtract or EvidenceItem (a Source → 422). [v0.1.2 · CR-I4-14]
limitations	text	Y	1	Known constraints and gaps.
alternative_explanations	text/ref[]	Y	1..n	Material alternatives.
confidence	confidence	Y	1	Confidence in judgment.
review_status	enum	Y	1	DRAFT, PEER_REVIEWED, APPROVED, SUPERSEDED [v0.1.1 · A09]
finalized	boolean	Y	1	Set by controlled finalization, with finalized_at, finalized_by and finalization_rationale. Finalized and reviewed are distinct states. [v0.1.2 · CR-I4-09]
review_required	boolean	Y	1	Set when a supporting Fact becomes DISPUTED or SUPERSEDED (Section 7.5). [v0.1.1 · A10] Also set for indirect dependencies (Section 7.5); a flagged draft cannot be finalized. [v0.1.2 · CR-I4-11]
review_trigger_ref	ref	N	0..1	VerificationDecision that triggered review_required. [v0.1.1 · A10] Since v0.1.3 it MAY also name the HandlingChange that upgraded an input (Section 16.4). [v0.1.3 · CR-I6-12]
review_reason	enum	N	0..1	Why review_required is set; registry enum review_reason: FACT_DISPUTED, FACT_SUPERSEDED, INPUT_HANDLING_UPGRADED. [v0.1.3 · CR-I6-04]
subject_refs	ref[]	N	0..n	Subjects of profiles and referrals (Entity, Asset), as the template requires. [v0.1.3 · CR-I6-01]
contact_point	text	N	0..1	Contact point of a referral package (template requirement). [v0.1.3 · CR-I6-01]
handling_decision_ref	ref	N	0..1	HandlingChange whose approved review downgraded the product; later inheritance floor checks accept the downgraded handling. [v0.1.3 · CR-I6-12]

Product versions and lifecycle [v0.1.3 · CR-I6-01, CR-I6-03, CR-I6-05]

- Submitting a product for review freezes a ProductVersion: an immutable snapshot with the rendered document (format csaml.product-document/1, including the evidence index) and its content_sha256. The first edit after a frozen version opens version n+1. [v0.1.3 · CR-I6-01]
- ProductVersion version_status (registry enum product_version_status): SUBMITTED → APPROVED / APPROVED_WITH_CHANGES / RETURNED / REJECTED → SUPERSEDED / RETRACTED. APPROVE sets the product APPROVED; APPROVE_WITH_CHANGES sets it REVIEWED, which is not disseminable until a new version is reviewed; RETURN and REJECT set it back to DRAFT. [v0.1.3 · CR-I6-03]

- A correction turns an APPROVED, REVIEWED or DISSEMINATED product into DRAFT version n+1 with a correction note; earlier versions remain valid until the correction is approved and then become SUPERSEDED, together with their disseminations. Withdrawal (case LEAD) sets the product WITHDRAWN, its versions RETRACTED, open reviews CANCELLED and disseminations REVOKED. [v0.1.3 · CR-I6-05]
- Clearing review_required is a review action: the approving reviewer sets it explicitly, or a REVIEW_FLAG_CLEARANCE review clears it. A DISSEMINATED product is never mutated; a CORRECTION review is opened instead. [v0.1.3 · CR-I6-04] | high_impact_adverse | boolean | N | 0..1 | Defaults to false. True when the assessment is adverse to a named person or organization or is otherwise designated high-impact by the case's review policy. [v0.1.1 · C10] | | disconfirming_searches | structured[] | C | 0..n | Recorded searches for information that would disconfirm the judgment. Each entry: searched_for (text), sources_consulted[] (1..n; each a source_ref and/or a description), result (text), rationale (text), recorded_by (principal), recorded_at (datetime). At least one entry is required before review approval when high_impact_adverse is true. Append-only. [v0.1.1 · C10] |

Normative rules:

- Assessment language SHALL distinguish known, assessed, and unknown information.
- An Assessment with high_impact_adverse = true SHALL NOT pass review (a Review with decision APPROVE or APPROVE_WITH_CHANGES), and neither SHALL an IntelligenceProduct that depends on it, unless the Assessment has at least one disconfirming_searches entry (SRS-FR-ASM-004). The check is enforced at review approval, not at finalization. Entries SHALL NOT be edited or deleted; a correction is a further entry. [v0.1.1 · C10]
- Assessment SHALL NOT imply criminal guilt beyond the available evidence and mandate.
- **Lifecycle (finalized vs reviewed).** Envelope status is DRAFT until finalization and FINALIZED afterwards; once a review records PEER_REVIEWED, APPROVED or SUPERSEDED, the envelope status follows review_status. Finalization freezes the content; review_status stays DRAFT until a review exists. [v0.1.2 · CR-I4-09]
- **Competing hypotheses.** Finalization SHALL be rejected (409 STATE_CONFLICT, reason COMPETING_HYPOTHESES_REQUIRED) when a Hypothesis in scope_refs has no competing hypothesis. [v0.1.2 · CR-I4-08]
- **Disconfirming searches after finalization.** Entries MAY be appended to a finalized Assessment until the Assessment (or a product depending on it) passes review approval; afterwards they are frozen (409). SRS-FR-ASM-004 is checked at review approval. This changes the I4 implementation, which froze searches at finalization. [v0.1.2 · CR-I4-10]
- **Revisions.** Every record_version of an Assessment SHALL be kept as an append-only revision snapshot. [v0.1.2 · CR-I4-05]

14. Confidence and Source Evaluation Model

CS-AML separates source reliability, information credibility, and analyst confidence. Implementations SHALL NOT compress these dimensions into a single unexplained score.

Dimension	Allowed values	Meaning
Source reliability	A-F	A highly reliable; B generally reliable; C mixed; D generally unreliable; E unreliable; F unknown
Information credibility	1-6	1 independently confirmed; 2 probably true; 3 possibly true; 4 doubtful; 5 improbable; 6 cannot be judged. Registry enum credibility_grade names these INDEPENDENTLY_CONFIRMED ... CANNOT_BE_JUDGED; the wire values remain the digit codes. [v0.1.2 · CR-I2-08]
Analyst confidence	HIGH / MODERATE / LOW, or INSUFFICIENT_BASIS (display: High / Moderate / Low / Insufficient basis)	Overall confidence in analytical judgment, based on evidence quality, independence, consistency, and remaining gaps. INSUFFICIENT_BASIS is not a level below LOW (Section 14.1). [v0.1.1 · A09]

14.1 Confidence

Structured confidence object embedded/referenced by analytical objects.

Field	Type	Req.	Cardinality	Semantics
level	enum	Y*	0..1	HIGH, MODERATE, LOW, INSUFFICIENT_BASIS; null only on drafts where no confidence judgement has yet been made. [v0.1.1 · A09]
basis	text	Y	1	Reason for confidence (rationale); mandatory for every level, including INSUFFICIENT_BASIS. [v0.1.1 · A09]
source_diversity	enum	N	0..1	SINGLE, MULTIPLE_RELATED, MULTIPLE_INDEPENDENT [v0.1.1 · A09]
material_gaps	ref[]	N	0..n	Gap references.
last_reviewed_at	datetime	Y	1	Review timestamp.
reviewer	principal	N	0..1	Reviewer if required.

Normative rules: [v0.1.1 · A09]

- INSUFFICIENT_BASIS means a judgement was attempted but the evidential basis is insufficient. It is NOT a level below LOW and SHALL NOT be converted to LOW, null, zero, or omitted in storage, APIs, exports, or visualizations.
- level = null is permitted only on draft objects where no confidence judgement has been made yet. A finalized (approved, established, or disseminated) analytical object SHALL carry a non-null level.
- No normalization, migration, or aggregation SHALL raise certainty (e.g. by mapping INSUFFICIENT_BASIS or null to any level, or LOW to MODERATE).

15. Intelligence Product and Dissemination Model

15.1 IntelligenceProduct

Represents an approved analytical output for an identified audience.

Field	Type	Req.	Cardinality	Semantics
product_type	enum	Y	1	FINANCIAL_INTELLIGENCE_NOTE, ENTITY_PROFILE, ASSET_PROFILE, NETWORK_ANALYSIS, REFERRAL_PACKAGE, CASE_REPORT, INVESTIGATION_BRIEF, PUBLICATION_SUPPORT. The first six are the MVP P0 templates (SRS-FR-PRD-001); INVESTIGATION_BRIEF and PUBLICATION_SUPPORT are not in MVP scope. [v0.1.1 · A06, A09]
title	string	Y	1	Neutral title.
assessment_refs	ref[]	Y	1..n	Underlying assessments.
evidence_manifest	ref[]	Y	1..n	Evidence inventory.
audience	enum/string[]	Y	1..n	Authorized audience.
classification	enum	Y	1	Sensitivity.
approval_state	enum	Y	1	DRAFT, REVIEWED, APPROVED, DISSEMINATED, WITHDRAWN [v0.1.1 · A09]
version	string	Y	1	Product version.
published_at	datetime	N	0..1	If published/disseminated.
review_required	boolean	Y	1	Set when a Fact underlying a referenced assessment becomes DISPUTED or SUPERSEDED (Section 7.5); a disseminated product is not mutated — a correction review task is created. [v0.1.1 · A10]
review_trigger_ref	ref	N	0..1	VerificationDecision that triggered review_required. [v0.1.1 · A10]

15.2 Review

Represents peer, legal, privacy, security, or red-team review.

Field	Type	Req.	Cardinality	Semantics
review_type	enum	Y	1	PEER, LEGAL, PRIVACY, SECURITY, RED_TEAM, EDITORIAL [v0.1.1 · A09]
target_ref	ref	Y	1	Object/product reviewed.
reviewer	principal	Y	1	Reviewer.
findings	text/structured	Y	1	Review findings.
decision	enum	Y	1	APPROVE, APPROVE_WITH_CHANGES, RETURN, REJECT [v0.1.1 · A09]

Field	Type	Req.	Cardinality	Semantics
completed_at	datetime	Y	1	Completion time.
review_kind	enum	Y	1	What the review decides; registry enum review_kind: PRODUCT_VERSION, ASSESSMENT, REVIEW_FLAG_CLEARANCE, HANDLING_CHANGE, CORRECTION. [v0.1.3 · CR-I6-02]
review_status	enum	Y	1	Registry enum review_status: OPEN, DECIDED, CANCELLED. reviewer, decision and completed_at are null while OPEN. [v0.1.3 · CR-I6-03]
target_version	string	N	0..1	Frozen product version under review (PRODUCT_VERSION reviews). [v0.1.3 · CR-I6-03]
requested_by	principal	Y	1	Principal who submitted or requested the review. [v0.1.3 · CR-I6-02]

Normative rules: [v0.1.3 · CR-I6-02]

- A Review targets a frozen product version, a finalized assessment, a review_required clearance, a handling change (classification downgrade or label removal) or a correction task. All kinds are decided through the same approve / request-changes / reject actions.
- Authors, contributors and the requester SHALL NOT decide a review (403; enforced by the database as well). Deciding requires a REVIEWER or LEAD case membership.

15.3 Dissemination

Records controlled release of a product or data package.

Field	Type	Req.	Cardinality	Semantics
product_ref	ref	Y	1	Product disseminated.
recipient	string/ref	Y	1	Recipient/authority/audience.
purpose	text	Y	1	Purpose.
released_at	datetime	Y	1	Release time.
released_by	principal	Y	1	Authorizer.
handling_caveats	string[]	N	0..n	Restrictions.
receipt_or_reference	string	N	0..1	Acknowledgement/reference.
product_version	string	Y	1	Frozen product version released. [v0.1.3 · CR-I6-06]
handling_classification	enum	Y	1	Handling classification of the release; at least the product's classification (defaults to it). [v0.1.3 · CR-I6-06]
dissemination_status	enum	Y	1	Registry enum dissemination_status: REQUESTED, APPROVED, REVOKED, SUPERSEDED. released_at / released_by are set on approval. [v0.1.3 · CR-I6-06]
package_scope_refs	ref[]	N	0..n	Complete approved scope of export packages. [v0.1.3 · CR-I6-06]
source_protected_release	boolean	Y	1	SOURCE_PROTECTED objects are in the approved scope only when true. [v0.1.3 · CR-I6-07]

Normative rules: [v0.1.3 · CR-I6-06, CR-I6-07]

- REQUESTED → APPROVED / REVOKED; APPROVED → REVOKED / SUPERSEDED. An approved release is immutable (database trigger). The approver is a REVIEWER or LEAD case member and never the requester.
- source_protected_release MAY be set only by an approver holding the per-case protected-source grant, with a rationale. Without it every SOURCE_PROTECTED object, index entry and citation is withheld from export packages; only the count is recorded.

15.4 ExportPackage and sharing log [v0.1.3 · CR-I6-08, CR-I6-09]

An ExportPackage is bound to one approved Dissemination and its product version. It records format (registry enum export_package_format: CSAML_PACKAGE_ZIP_V1), package_status (registry enum export_package_status: QUEUED, GENERATING, READY, FAILED), the requested object_refs, the files with their SHA-256, package_sha256, manifest_sha256 and the redactions (counts and reasons, never content). Every download re-checks authorization and approval validity. Sharing-log entries are immutable and carry entry_type (registry enum sharing_log_entry_type: RELEASE_APPROVED, EXPORT_GENERATED, RELEASE_REVOKED), product version, handling caveats and handling classification. Package generation runs as a Job (registry enum job_status: QUEUED, RUNNING, SUCCEEDED, FAILED).

16. Privacy, Classification, and Access-Control Metadata

This section defines the authoritative CS-AML information-classification model. All other CS-AML documents SHALL use these levels and wire values. [v0.1.1 · A08]

Levels are ordered from least to most restrictive:

Classification (wire value)	Display label	Meaning
PUBLIC	Public	Suitable for public release after normal review.
INTERNAL	Internal	Routine internal operational information.
SENSITIVE	Sensitive	Could create privacy, reputational, safety, or investigative harm if disclosed.
RESTRICTED	Restricted	High-risk information requiring named-role or case-specific authorization.
SOURCE_PROTECTED	Source-protected	Information whose disclosure could identify or endanger a confidential source. [v0.1.1 · A08]

v0.1 wrote the last level as SOURCE-PROTECTED; the v0.1.1 wire value is SOURCE_PROTECTED. [v0.1.1 · A08, A09]

Access labels MAY add purpose, jurisdiction, source-protection, embargo, legal-review, or compartment restrictions. Implementations SHALL enforce the most restrictive applicable label.

Normative rules: [v0.1.1 · A08]

- Access labels are additive restrictions on top of the classification level. Access requires satisfying the object's classification level and every applicable access label; neither replaces the other.
- Derived objects and exports SHALL inherit the highest classification (and the union of access labels) of their inputs, unless a recorded reviewer downgrade decision exists for that object.
- An unknown, missing, or unrecognized classification value SHALL fail closed: access is denied and the object is flagged for classification. It SHALL NOT default to a less restrictive level.
- **Inheritance is enforced, not silently raised.** A declared classification below the maximum of the inputs, or access labels missing an input label, SHALL be rejected (422); omitted values default to the inherited ones. PATCH MAY only upgrade or add labels; downgrades SHALL be rejected (403) until a recorded reviewer downgrade decision exists. When an input is later upgraded, dependent derived objects SHALL be flagged "re-review required"; their classification SHALL NOT be raised automatically. [v0.1.2 · CR-I2-05]
- Case classification and labels are changed only by the case LEAD, upgrades only (Section 6.1). [v0.1.2 · CR-I1-09]
- The "recorded reviewer downgrade decision" is a HandlingChange with direction DOWNGRADE that names its approved HANDLING_CHANGE review (Section 16.4). [v0.1.3 · CR-I6-12]

Principal clearance model [v0.1.2 · CR-I1-10]

A principal's clearance is represented by identity-provider (Keycloak realm) roles; see also Technical Stack v0.1.4 §10. [v0.1.2 · CR-I1-10]

IdP role	Effect
(none)	Baseline clearance INTERNAL: PUBLIC and INTERNAL objects of the principal's cases.
csaml-clearance-sensitive	Clearance up to SENSITIVE.
csaml-clearance-restricted	Clearance up to RESTRICTED.

IdP role	Effect
csaml-protected-source	Eligibility for SOURCE_PROTECTED material; effective only together with the per-case grant protected_source_authorized on the CaseMembership (Section 6.3).
csaml-label-	Satisfies the access label ; every label of an object must be satisfied.

Clearance never replaces case membership: access requires membership, sufficient clearance and every access label. Unknown labels fail closed. [v0.1.2 · CR-11-10]

Legacy mapping from the Framework v0.1 four-level scheme [v0.1.1 · A08]

The parent Framework v0.1 Expanded (Section 6.4) used Public / Internal / Restricted / Highly Restricted. Records migrated from that scheme SHALL be mapped as follows. The last two rows SHALL NOT be mapped automatically.

Framework v0.1 label	v0.1.1 level
Public	PUBLIC
Internal	INTERNAL
Restricted	SENSITIVE or RESTRICTED — chosen by the data owner during migration; RESTRICTED until decided (fail closed).
Highly Restricted	SOURCE_PROTECTED only when the reason is source-identifying information; otherwise RESTRICTED plus the relevant access label (e.g. legal-privilege, physical-security). Never auto-map.

16.1 RetentionRule

Represents retention and disposition policy attached to objects or classes.

Field	Type	Req.	Cardinality	Semantics
rule_id	string	Y	1	Policy identifier.
applies_to	enum/ref	Y	1	Object type/object. Registry enum retention_target_type: EVIDENCE_ITEM, EXPORT_PACKAGE, CASE (MVP 0.1). Never changes after creation. [v0.1.3 · CR-17-02]
applies_to_classifications	enum[]	N	0..n	Narrows the rule to objects of these classifications (SRS-FR-ADM-003 "by object type/classification"); empty = all. [v0.1.3 · CR-17-02]
retention_period	duration/date	Y	1	Retention term.
trigger	enum	Y	1	CREATION, CASE_CLOSURE, DISSEMINATION, LEGAL_HOLD_RELEASE, OTHER [v0.1.1 · A09]
disposition	enum	Y	1	DELETE, ANONYMIZE, ARCHIVE, REVIEW [v0.1.1 · A09]
legal_hold	boolean	Y	1	Whether disposition suspended.
rule_status	enum	Y	1	Registry enum retention_rule_status: ACTIVE, RETIRED (final; rules are never deleted). [v0.1.3 · CR-17-03]

Normative rules: [v0.1.3 · CR-17-01, CR-17-05, CR-17-06]

- Rules are versioned by record_version and never deleted; every change is audited.
- When several active rules match a target, the most protective governs: the target is eligible only when every matching period has expired, and the latest date wins. CASE_CLOSURE uses Case.closed_at (Section 6.1).
- Evaluation only reports or proposes; nothing is disposed of automatically. Disposition requires an approved DispositionRecord (Section 16.3).
- MVP 0.1 executes DELETE (evidence originals and export packages: the stored bytes are purged and the record stays as the tombstone), ARCHIVE and REVIEW (recorded retention state). ANONYMIZE and whole-case DELETE SHALL be refused (422) in MVP 0.1; their semantics (a reviewed anonymiser, a case-level tombstone) are deferred to v0.2. [v0.1.3 · CR-17-05]

16.2 LegalHold [v0.1.3 · CR-17-03]

Suspends disposition of a case (and every object linked to it) or of one object. Required as the legal hold record of CIG PRI-02.

Field	Type	Req.	Cardinality	Semantics
target_type	enum	Y	1	Registry enum retention_target_type.
target_ref	ref	Y	1	Held case or object.
reason	text	Y	1	Why the hold is placed.
authority_reference	string	N	0..1	Order, request or authority reference.
hold_status	enum	Y	1	Registry enum legal_hold_status: ACTIVE, RELEASED.
placed_by / placed_at	principal / datetime	Y	1	Placed by a platform administrator or the case LEAD.
released_by / released_at / release_reason	principal / datetime / text	N	0..1	Recorded once, on release.

Normative rules:

- A legal hold is never deleted; only its release is recorded, once. While a hold is ACTIVE, approval and execution of any disposition of the held target SHALL be refused (409; also enforced by the database).

16.3 DispositionRecord [v0.1.3 · CR-17-04]

The disposition log required by CIG PRI-02: one record per proposed disposition of one target.

Field	Type	Req.	Cardinality	Semantics
rule_ref / rule_version	ref / integer	Y	1	Governing RetentionRule and its version.
target_type / target_ref	enum / ref	Y	1	Target (registry enum retention_target_type).
disposition / trigger	enum	Y	1	From the rule (retention_disposition, retention_trigger).
trigger_at / eligible_at	datetime	Y	1	Trigger time and the date the target became eligible.
disposition_status	enum	Y	1	Registry enum disposition_status: PROPOSED, APPROVED, REJECTED, EXECUTING, EXECUTED.
proposed_by / proposed_at	principal / datetime	Y	1	Proposer (the evaluation).
decided_by / decided_at / decision_rationale	principal / datetime / text	N	0..1	Decision.
executed_by / executed_at / outcome	principal / datetime / structured	N	0..1	Execution and its outcome (e.g. purged versions, retained hash and size).

Normative rules:

- PROPOSED → APPROVED / REJECTED → EXECUTING → EXECUTED. The decider is the LEAD of every case the target is linked to and never the proposer (403; database CHECK). A target linked to no case cannot be approved (fail closed).
- Execution re-checks the rule and legal holds and commits in two steps (EXECUTING, then EXECUTED) so that purged bytes are never without a record. Terminal records are frozen. Records and audit events are never deleted.
- A record carries the target's classification, access labels and case links and is visible exactly where the target is.

16.4 HandlingChange [v0.1.3 · CR-16-12]

Append-only record of a classification or access-label change of a canonical object.

Field	Type	Req.	Cardinality	Semantics
target_type / target_ref	string / ref	Y	1	Changed object.
from_classification / to_classification	enum	Y	1	Classification before and after.
from_labels / to_labels	string[]	Y	0..n	Access labels before and after.
direction	enum	Y	1	Registry enum handling_change_direction: UPGRADE, DOWNGRADE.

Field	Type	Req.	Cardinality	Semantics
review_ref	ref	C	0..1	Approved HANDLING_CHANGE review; required for DOWNGRADE (database CHECK).
flagged_refs	ref[]	N	0..n	Derived objects flagged review_required by an upgrade (Section 16, CR-I2-05).
changed_by / changed_at	principal / datetime	Y	1	Actor and time.

Normative rules:

- UPGRADE is applied directly; DOWNGRADE (including label removal) only with its approved review. Records are never updated or deleted.
- review_trigger_ref of a flagged assessment or product MAY name the HandlingChange; a downgraded product keeps it as handling_decision_ref (Section 15.1).

17. Temporal, Versioning, and Audit Semantics

CS-AML distinguishes system time (when the system knew/recorded something) from valid time (when the represented fact/relationship was true in the real world). Where historically material, both SHOULD be stored.

valid_from / valid_to	= real-world validity
created_at / updated_at	= system record time
record_version	= object mutation sequence
AuditEvent	= who changed what, when, and why

17.1 AuditEvent

Append-only record of material actions.

Field	Type	Req.	Cardinality	Semantics
actor	principal	Y	1	Human/service principal.
action	enum/string	Y	1	CREATE, UPDATE, MERGE, SPLIT, REVIEW, APPROVE, DISSEMINATE, RESTRICT, DELETE, EXPORT, LOGIN_SENSITIVE, OTHER [v0.1.1 · A09]
target_ref	ref	Y	1	Affected object.
occurred_at	datetime	Y	1	Time.
reason	text	N	0..1	Reason/context.
before_hash	string	N	0..1	Optional integrity reference.
after_hash	string	N	0..1	Optional integrity reference.
case_context	ref	N	0..1	Case context if applicable.

Normative rules:

- Audit events SHALL be append-only for normal users.
- Administrative correction of audit data SHALL itself generate an audit event.

18. Cross-Object Integrity Rules

Rule	Constraint
DM-I01	Every Fact SHALL reference at least one EvidenceItem or EvidenceExtract (supporting_evidence 1..n); supporting Claims are optional (0..n) and never replace evidence. Every Fact SHALL have the CREATE VerificationDecision written in the same transaction. [v0.1.1 · C01] [v0.1.1 · C02]
DM-I02	Every Indicator SHALL reference supporting evidence and at least one subject object.
DM-I03	Every TypologyMatch SHALL reference one catalogue typology and one or more indicators.
DM-I04	Every Assessment SHALL reference supporting analytical objects and include limitations.
DM-I05	A ValueFlow with flow_class=DIRECT SHALL NOT exist without direct evidence of movement.

Rule	Constraint
DM-I06	A Relationship of type OWNS/BENEFICIAL_OWNER_OF/CONTROLS SHALL include confidence with a non-empty confidence.basis (the basis of the relationship); the Relationship has no separate basis field. Detailed ownership/control basis is held on OwnershipInterest.basis / ControlAssertion.control_basis. <i>[v0.1.2 · CR-I2-15]</i>
DM-I07	Entity merge SHALL preserve precursor identifiers and generate AuditEvent. Every merge and unmerge SHALL be recorded as a ResolutionDecision. <i>[v0.1.1 · ER]</i>
DM-I08	External dissemination SHALL reference an approved IntelligenceProduct.
DM-I09	Source-protected information (classification SOURCE_PROTECTED or a source-protection access label) SHALL NOT be exported into lower-classification products without explicit de-identification review. <i>[v0.1.1 · A08]</i>
DM-I10	Deletion/anonymization SHALL respect evidence preservation obligations and legal holds.
DM-I11	Every Fact SHALL reference at least one VerificationDecision; a Claim SHALL NOT be stored, shown, or exported as a Fact without one. <i>[v0.1.1 · A10]</i>
DM-I12	A Fact with fact_status = SUPERSEDED SHALL reference its replacement Fact (superseded_by). <i>[v0.1.1 · A10]</i>
DM-I13	An object with missing or unrecognized classification SHALL be treated as inaccessible (fail closed) and flagged for classification. <i>[v0.1.1 · A08]</i>
DM-I14	A derived object or export SHALL carry a classification at least as restrictive as the highest classification of its inputs, unless a recorded reviewer downgrade decision exists. <i>[v0.1.1 · A08]</i>
DM-I15	Entity.resolution_status SHALL be derivable from the Entity's ResolutionDecision history; a resolution_status change without a corresponding ResolutionDecision SHALL be rejected. <i>[v0.1.1 · ER]</i>

19. Canonical Graph Mapping

Graph implementations SHOULD use Entity, Asset, Event, and selected contextual objects as nodes, with Relationship and ValueFlow as first-class edges or edge-objects. Evidence SHALL NOT be lost when graph projections are materialized.

```
(:Person)-[:DIRECTOR_OF {relationship_id,...}]->(:Organization)
(:Person)-[:BENEFICIAL_OWNER_OF {ownership_interest_id,...}]->(:Organization)
(:Organization)-[:AWARDED_CONTRACT]->(:Contract)
(:Organization)-[:ACQUIRED]->(:Asset)
(:Entity)-[:VALUE_FLOW {value_flow_id, flow_class,...}]->(:Entity)
```

19.1 Graph projection rule

A graph edge is a projection of a canonical Relationship or ValueFlow object. The canonical object remains authoritative because it contains provenance, validity period, confidence, and handling metadata that may exceed native graph-edge capabilities.

20. Reference Relational Mapping

```
cases
case_memberships          (v0.1.2 · CR-I1-02)
investigation_questions
charter_versions
case_gates
case_gate_decisions
case_tasks
case_task_events
sources
evidence_items
evidence_extracts
derived_texts
derived_text_pages
evidence_integrity_checks (append-only verification results)
upload_sessions
```

```

claims
verification_decisions    (v0.1.1 · A10)
facts
entities
entity_identifiers
resolution_decisions      (v0.1.1 · ER)
relationships
ownership_interests
control_assertions
assets
events
value_flows
value_flow_legs
indicators
typology_matches
hypotheses
hypothesis_links
intelligence_gaps
assessments
assessment_disconfirming_searches
assessment_revisions      (v0.1.2 · CR-I4-05)
intelligence_products
product_versions
reviews
disseminations
export_packages
sharing_log_entries
jobs
handling_changes
retention_rules
legal_holds
disposition_records
search_documents
audit_events

```

v0.1.2 adds `case_memberships`, `upload_sessions`, `evidence_integrity_checks`, `assessment_disconfirming_searches` and `assessment_revisions` to the list; `hypothesis_links` holds the append-only matrix cells (Section 13.1). *[v0.1.2 · CR-I1-02, CR-I4-04, CR-I4-05]*

v0.1.3 adds `product_versions`, `export_packages`, `sharing_log_entries`, `jobs`, `handling_changes`, `retention_rules`, `legal_holds` and `disposition_records`; `search_documents` is a derived, rebuildable search projection, never canonical. *[v0.1.3 · CR-I6-01, CR-I6-08, CR-I6-09, CR-I6-12, CR-I7-01, CR-I7-03, CR-I7-04, CR-I5-01]*

v0.1.4 adds `investigation_questions`, `charter_versions`, `case_gates`, `case_gate_decisions` (append-only), `case_tasks`, `case_task_events` (append-only), `derived_texts` and `derived_text_pages` (append-only). *[v0.1.4 · CR-N-01, CR-N-03, CR-N-04, CR-N-09]*

This list is illustrative. Implementations MAY normalize or denormalize differently provided semantic and integrity requirements are preserved.

21. Canonical JSON Example

Enumeration values in this example use the UPPER_SNAKE_CASE wire form (Section 4.1). *[v0.1.1 · A09]*

```

{
  "id": "vf-7c2e...",
  "object_type": "ValueFlow",
  "schema_version": "0.1",
  "flow_class": "RECONSTRUCTED",
  "flow_type": "ASSET_PURCHASE",
  "origin_ref": "org-I23",
  "destination_ref": "asset-456",
  "amount": {
    "value": 8500000000,
    "currency": "IDR",
    "precision": "ESTIMATED"
  },
  "occurred_at": {

```

```

    "from": "2026-05-01",
    "to": "2026-05-31"
  },
  "evidence_refs": [
    "evd-11",
    "evd-19"
  ],
  "reconstruction_basis": "Contract award and subsequent deed records indicate a plausible economic link; no bank
  ↳ settlement record is available.",
  "confidence": {
    "level": "MODERATE",
    "basis": "Two independent documentary sources; settlement route unknown."
  },
  "classification": "SENSITIVE",
  "record_version": 3
}

```

22. API and Exchange Requirements

- APIs SHALL expose object IDs, schema version, record version, classification, and provenance references.
- APIs SHOULD support field-level filtering to prevent unnecessary disclosure of sensitive attributes.
- Bulk exports SHALL preserve relationship/evidence IDs so provenance survives movement between systems.
- Systems SHALL reject writes that violate mandatory cross-object integrity rules.
- Machine-generated analytical suggestions SHALL be marked with `creator_type = AUTOMATION` or `AI` and SHALL require human review before becoming an Assessment. *[v0.1.1 · A09]*

23. Interoperability and External Identifiers

The canonical model is implementation-neutral and MAY map to external ontologies or identifiers. External IDs SHALL be stored as typed identifiers and SHALL NOT replace internal canonical IDs.

Domain	Examples	Model treatment
Corporate	registration number, LEI, registry ID	Entity.identifier[]
Persons	official registry identifiers where lawful	Entity.identifier[] with restricted handling where sensitive
Sanctions/PEP	list-specific subject IDs	External identifier + Source/Evidence
Crypto	wallet/address	Entity type=wallet; chain/network as typed attribute
Procurement	tender/contract IDs	Contract Entity/Event/Source refs
Geospatial	parcel/registry identifiers	Asset/Location identifiers

24. Schema Versioning and Migration

- Every stored object SHALL declare `schema_version`.
- Breaking semantic changes SHALL increment the major/minor model version according to framework governance.
- Migrations SHALL preserve provenance and audit history.
- A migration SHALL NOT silently convert reconstructed value flow into direct value flow, or claim into fact, or hypothesis into assessment.
- Deprecated fields SHOULD remain readable for a defined transition period.

25. Data Quality and Validation

Quality dimension	Expected measure
Completeness	Required fields populated and required links present.
Provenance coverage	Share of facts/indicators/assessments traceable to evidence.
Entity resolution quality	Merge/split error rate; unresolved duplicate rate.
Temporal quality	Coverage of validity dates for time-sensitive relationships.
Confidence hygiene	Analytical objects with explicit confidence basis.
Classification hygiene	Sensitive objects carrying appropriate handling labels.
Orphan rate	Objects lacking expected parent/context/evidence links.
Audit completeness	Material actions represented by AuditEvent.

26. Data Model Conformance

An implementation claiming to implement CS-AML Data Model v0.1.4 SHALL demonstrate the following minimum capabilities:

- Persistent canonical IDs and schema versioning.
- Separate Source, Evidence, Fact, Indicator, Hypothesis, Assessment, and IntelligenceProduct objects.
- Reusable Entity objects with evidence-based resolution history.
- Evidence-linked Relationship and ValueFlow representations.
- Explicit value-flow epistemic class.
- Confidence and intelligence-gap representation.
- Sensitivity/classification metadata and access enforcement.
- Append-only audit history for material actions.
- Export that preserves provenance references and object identity.
- Validation of the mandatory cross-object integrity rules in Section 18.

No implementation has yet been assessed against these capabilities; they are targets, not evidence of conformance. [v0.1.1 · A01]

Conformance principle

A schema is not conformant merely because it uses similar table or class names. Conformance depends on preserving the semantics, separations, provenance requirements, and integrity constraints defined by this specification.

Annex A. Controlled Enumerations (Baseline)

Annex A, together with the enumerations stated in the field tables of this specification, is the controlled-enumeration registry for CS-AML. All wire values are UPPER_SNAKE_CASE (Section 4.1); display labels are separate and translatable. Enumeration lists in other CS-AML documents derive from this registry. [v0.1.1 · A09]

Enumeration	Baseline values
entity_type	PERSON, ORGANIZATION, ACCOUNT, ADDRESS, DOMAIN, PHONE, EMAIL, WALLET, PROPERTY, VEHICLE, VESSEL, AIRCRAFT, CONTRACT, PROJECT, OTHER [v0.1.1 · A09]
flow_class	DIRECT, DOCUMENTED, RECONSTRUCTED, HYPOTHETICAL [v0.1.1 · A09]
confidence.level	HIGH, MODERATE, LOW, INSUFFICIENT_BASIS (null only on drafts with no judgement yet; INSUFFICIENT_BASIS is not a level below LOW — Section 14.1) [v0.1.1 · A09]

Enumeration	Baseline values
classification	PUBLIC, INTERNAL, SENSITIVE, RESTRICTED, SOURCE_PROTECTED (ordered least → most restrictive; unknown/missing fails closed — Section 16) [v0.1.1 · A08, A09]
hypothesis.status	OPEN, SUPPORTED, WEAKENED, REJECTED, INCONCLUSIVE [v0.1.1 · A09]
typology_match.consistency_level	NO_BASIS, WEAK, PLAUSIBLE, STRONG, COMPELLING [v0.1.1 · A09]
review.decision	APPROVE, APPROVE_WITH_CHANGES, RETURN, REJECT [v0.1.1 · A09]
claim.claim_status	RECORDED, UNDER_REVIEW, CORROBORATED, CONTRADICTED, UNRESOLVED [v0.1.1 · A10]
fact.fact_status	PROVISIONAL, ESTABLISHED, DISPUTED, SUPERSEDED [v0.1.1 · A10]
verification_decision.decision	Claims: UNDER_REVIEW, CORROBORATED, CONTRADICTED, UNRESOLVED; Facts: CREATE (label "Create fact"), ESTABLISH, DISPUTE, SUPERSEDE [v0.1.1 · A10]
entity.resolution_status	UNRESOLVED, RESOLVED, CONFLICTED, MERGED, SPLIT (state only; changed by ResolutionDecision) [v0.1.1 · A09, ER]
resolution_decision.decision	MERGE, KEEP_SEPARATE, POSSIBLE_MATCH, DEFER, UNMERGE [v0.1.1 · ER]
case_membership.role	LEAD, ANALYST, REVIEWER [v0.1.2 · CR-I1-02]
case.risk_rating	LOW, MEDIUM, HIGH, CRITICAL (ordered; minimal set) [v0.1.2 · CR-I1-08]
case.closure_reason	OBJECTIVES_MET, INSUFFICIENT_BASIS_TO_CONTINUE, REFERRED, OUT_OF_SCOPE, DUPLICATE, LEGAL_OR_SAFETY_CONSTRAINT, OTHER (minimal set) [v0.1.2 · CR-I1-08]
upload_session.status	INITIATED, CONTENT_RECEIVED, COMPLETED (expiry is expressed by expires_at) [v0.1.2 · CR-I2-02]
envelope.status	REGISTERED, RECORDED, DRAFT, FINALIZED (for classes without their own lifecycle: REGISTERED for Source, EvidenceExtract, Asset; RECORDED for Event, ValueFlow, TypologyMatch; DRAFT / FINALIZED for Assessment before a review outcome) [v0.1.2 · CR-I2-03, CR-I3-05, CR-I4-09]
claim.credibility_grade	INDEPENDENTLY_CONFIRMED, PROBABLY_TRUE, POSSIBLY_TRUE, DOUBTFUL, IMPROBABLE, CANNOT_BE_JUDGED (wire codes "1"... "6", registry exception) [v0.1.2 · CR-I2-08]
money.precision	EXACT, APPROXIMATE, ESTIMATED (a range is never exact) [v0.1.2 · CR-I3-06]
temporal_value.precision	DATETIME, DAY, MONTH, YEAR, RANGE, UNKNOWN (string shape must match) [v0.1.2 · CR-I3-06]
hypothesis.role	PRINCIPAL, ALTERNATIVE_LEGITIMATE, ALTERNATIVE_MECHANISM, INSUFFICIENT_INFORMATION [v0.1.2 · CR-I4-06]
hypothesis_link.effect	SUPPORTS, CONTRADICTS, NEUTRAL, UNKNOWN [v0.1.2 · CR-I4-07]
search_hit.object_type	CASE, ENTITY, SOURCE, EVIDENCE_ITEM, EVIDENCE_EXTRACT, CLAIM, FACT, RELATIONSHIP, ASSET, EVENT, VALUE_FLOW, PRODUCT [v0.1.3 · CR-I5-03], DERIVED_TEXT [v0.1.4 · CR-N-11]
search_hit.epistemic_status	CLAIM, FACT, EXCERPT, RECORD (a claim is never shown as a fact) [v0.1.3 · CR-I5-03], DERIVED (never the original) [v0.1.4 · CR-N-11]
graph_lead.lead_type	SHARED_IDENTIFIER, SHARED_PHONE, SHARED_EMAIL, SHARED_ADDRESS (candidates, never facts or edges) [v0.1.3 · CR-I5-05]
graph_query.truncation_reason	NODE_BUDGET, EDGE_BUDGET, TIME_BUDGET, LEAD_BUDGET [v0.1.3 · CR-I5-06]
review.review_kind	PRODUCT_VERSION, ASSESSMENT, REVIEW_FLAG_CLEARANCE, HANDLING_CHANGE, CORRECTION [v0.1.3 · CR-I6-02]
review.review_status	OPEN, DECIDED, CANCELLED [v0.1.3 · CR-I6-03]
product_version.version_status	SUBMITTED, APPROVED, APPROVED_WITH_CHANGES, RETURNED, REJECTED, SUPERSEDED, RETRACTED [v0.1.3 · CR-I6-03]
intelligence_product.review_reason	FACT_DISPUTED, FACT_SUPERSEDED, INPUT_HANDLING_UPGRADED [v0.1.3 · CR-I6-04]
dissemination.dissemination_status	REQUESTED, APPROVED, REVOKED, SUPERSEDED [v0.1.3 · CR-I6-06]

Enumeration	Baseline values
export_package.format	CSAML_PACKAGE_ZIP_V1 [v0.1.3 · CR-16-08]
export_package.package_status	QUEUED, GENERATING, READY, FAILED [v0.1.3 · CR-16-08]
sharing_log.entry_type	RELEASE_APPROVED, EXPORT_GENERATED, RELEASE_REVOKED [v0.1.3 · CR-16-08]
job.status	QUEUED, RUNNING, SUCCEEDED, FAILED [v0.1.3 · CR-16-09]
handling_change.direction	UPGRADE, DOWNGRADE [v0.1.3 · CR-16-12]
retention_rule.applies_to	EVIDENCE_ITEM, EXPORT_PACKAGE, CASE (MVP 0.1) [v0.1.3 · CR-17-02]
retention_rule.rule_status	ACTIVE, RETIRED [v0.1.3 · CR-17-03]
legal_hold.hold_status	ACTIVE, RELEASED [v0.1.3 · CR-17-03]
disposition_record.disposition_status	PROPOSED, APPROVED, REJECTED, EXECUTING, EXECUTED [v0.1.3 · CR-17-04]
investigation_question.priority	PRIMARY, SECONDARY [v0.1.4 · CR-N-02]
gate.gate_status	NOT_SUBMITTED, SUBMITTED, APPROVED, REJECTED [v0.1.4 · CR-N-03]
task.task_type	COLLECTION, VERIFICATION, ANALYSIS, REVIEW, OTHER [v0.1.4 · CR-N-04]
task.task_status	OPEN, IN_PROGRESS, BLOCKED, DONE, CANCELLED (DONE and CANCELLED final) [v0.1.4 · CR-N-04]
derived_text.transformation	OCR, TEXT_EXTRACTION, MACHINE_TRANSLATION [v0.1.4 · CR-N-09, CR-N-10]
derived_text.status	QUEUED, PROCESSING, REVIEW_REQUIRED, REVIEWED, FAILED [v0.1.4 · CR-N-09]
derived_text_page.method	OCR, TEXT_LAYER, PLAIN_TEXT, MACHINE_TRANSLATION [v0.1.4 · CR-N-09, CR-N-10]
evidence_item.derivation_type	OCR_TEXT, TEXT_EXTRACTION, MACHINE_TRANSLATION (open vocabulary; other values for registered derivatives) [v0.1.4 · CR-N-09]
job.job_type	EXPORT_PACKAGE, TEXT_EXTRACTION, MACHINE_TRANSLATION [v0.1.4 · CR-N-12]

Annex B. Canonical Relationship Vocabulary (Baseline)

Relationship	Direction	Semantics
OWNS	Directed	Legal ownership unless subtype says otherwise.
BENEFICIAL_OWNER_OF	Directed	Beneficial ownership/economic interest.
CONTROLS	Directed	Control without requiring ownership.
DIRECTOR_OF	Directed	Corporate role; from Person to Organization (Person --DIRECTOR_OF-> Organization). [v0.1.1 · A14]
OFFICER_OF	Directed	Organizational role.
EMPLOYED_BY	Directed	Employment/role.
RELATED_TO	Symmetric	Familial/known relation; subtype required where possible.
ASSOCIATE_OF	Symmetric	Documented association; SHALL NOT imply wrongdoing.
SHARES_ADDRESS_WITH	Symmetric	Common address.
SHARES_PHONE_WITH	Symmetric	Common phone.
SHARES_DEVICE_WITH	Symmetric	Common device if lawfully obtained.
CONTRACTED_BY	Directed	Contract relationship.
SUPPLIER_TO	Directed	Supply relationship.
ACQUIRED	Directed	Entity acquired asset.
SOLD_TO	Directed	Asset transfer.
REPRESENTED_BY	Directed	Professional/legal representation.
PAID_BY	Directed	Documented payer relationship, not necessarily direct bank proof.
AUTHORIZED_SIGNATORY_OF	Directed	Signing authority; from Person to Organization or Account. [v0.1.1 · C09]
COMMISSIONER_OF	Directed	Supervisory-board (commissioner) role; from Person to Organization. [v0.1.1 · C09]
MANAGES	Directed	Day-to-day management of an entity or asset without implying ownership or control. [v0.1.1 · C09]

Relationship	Direction	Semantics
USES	Directed	Entity uses or occupies an asset without documented ownership. <i>[v0.1.1 · C09]</i>
LENDER_TO	Directed	Lending relationship; from lender to borrower. <i>[v0.1.1 · C09]</i>
LEASED_TO	Directed	Lease of an asset; from lessor to lessee. <i>[v0.1.1 · C09]</i>
DONATED_TO	Directed	Documented donation; from donor to recipient. <i>[v0.1.1 · C09]</i>
FUNDED_BY	Directed	Documented funding; from funded entity to funder. <i>[v0.1.1 · C09]</i>
SHARES_DOMAIN_WITH	Symmetric	Common internet domain or email domain. <i>[v0.1.1 · C09]</i>
TRANSFERRED_VALUE_TO	Directed	Value transfer from sender to receiver; requires evidence or clearly marked reconstruction; prefer ValueFlow where movement semantics are material. <i>[v0.1.1 · C09]</i>

Display synonyms mapped to registered values. *[v0.1.1 · C09]* Relationship names used in the Investigation Methodology §13, the Information Architecture §10 and the Framework Expanded §11.2 that have an exact registered equivalent are display synonyms only and SHALL NOT be stored or exchanged as wire values. "Inverse" means the edge is stored with the registered type and the endpoints swapped.

Name used in a document	Stored as (wire value)
SHAREHOLDER_OF	OWNS, with an OwnershipInterest (Section 9.2) recording ownership_type and percentage
OWNS_ASSET	OWNS (to_entity is the Asset)
ACQUIRED_FROM	Inverse of SOLD_TO (seller --SOLD_TO--> buyer); ACQUIRED for the buyer-to-asset edge
BORROWER_FROM, LOANED_TO	LENDER_TO (BORROWER_FROM is the inverse)
USES_ASSET	USES
RELATIVE_OF	RELATED_TO, with a familial subtype
REPRESENTS, ACTS_FOR	Inverse of REPRESENTED_BY (principal --REPRESENTED_BY--> agent); nominee arrangements use OwnershipInterest NOMINEE_ASSERTED or ControlAssertion
SHARES_CONTACT_WITH	SHARES_PHONE_WITH or SHARES_ADDRESS_WITH; shared email or web domain uses SHARES_DOMAIN_WITH
TRANSFERRED_TO	TRANSFERRED_VALUE_TO
RECEIVED_VALUE_FROM	Inverse of TRANSFERRED_VALUE_TO
CONTROLS_ASSET	CONTROLS (to_entity is the Asset)
BUSINESS_PARTNER_OF	ASSOCIATE_OF, with a business-partner subtype
SUBCONTRACTED_TO	Inverse of CONTRACTED_BY (subcontractor --CONTRACTED_BY--> contractor)
INVESTED_IN	OWNS with an OwnershipInterest (ECONOMIC_INTEREST), or LENDER_TO for debt
NOMINEE_FOR	Not a relationship type: OwnershipInterest with ownership_type NOMINEE_ASSERTED, or a ControlAssertion

Annex C. Minimal Implementation Object Set

A minimum viable conformant implementation SHOULD support at least the following persistent objects:

Case	
Source	
EvidenceItem	
EvidenceExtract	
Claim	(v0.1.1 · A10)
VerificationDecision	(v0.1.1 · A10)
Fact	
Entity	
ResolutionDecision	(v0.1.1 · ER)
Relationship	

```

Asset
Event
ValueFlow
Indicator
TypologyMatch
Hypothesis
IntelligenceGap
Assessment
IntelligenceProduct
Review
Dissemination
AuditEvent

```

Annex D. Example Analytical Chain

The example below is drawn as a single vertical chain. Relationships are written from --TYPE--> to; the Event is a separate node and is not an endpoint of the relationship. [v0.1.1 · A14]

```

SOURCE
  Government procurement portal
  |
  v
EVIDENCE
  Contract award record
  |
  v
FACT
  Company A received Contract X on 12 May
  |
  v
EVENT
  Contract award (participant: Company A; object: Contract X; date: 12 May)
  |
  v
RELATIONSHIP
  Person B --DIRECTOR_OF--> Company A
  |
  v
INDICATOR
  Related entity acquired property shortly after award
  |
  v
HYPOTHESIS
  Value may have been diverted through related entity
  |
  v
VALUE FLOW (RECONSTRUCTED)
  Contract -> Company A -> related entity -> asset
  |
  v
ASSESSMENT
  Pattern is consistent with procurement-value diversion;
  settlement route remains unknown.
  |
  v
INTELLIGENCE PRODUCT

```

Note: DIRECTOR_OF is directed from Person to Organization (Section 19; Annex B). The v0.1 drawing showed Company A --DIRECTOR_OF--> Person B, which reversed the edge; that was an error in the example, not a change in the relationship vocabulary. [v0.1.1 · A14]